

Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 270 factors, ordered by in-sample L-S CAGR (last 48 months, net of cost).

HOW TO READ

■ Long ■ Short ■ Neutral

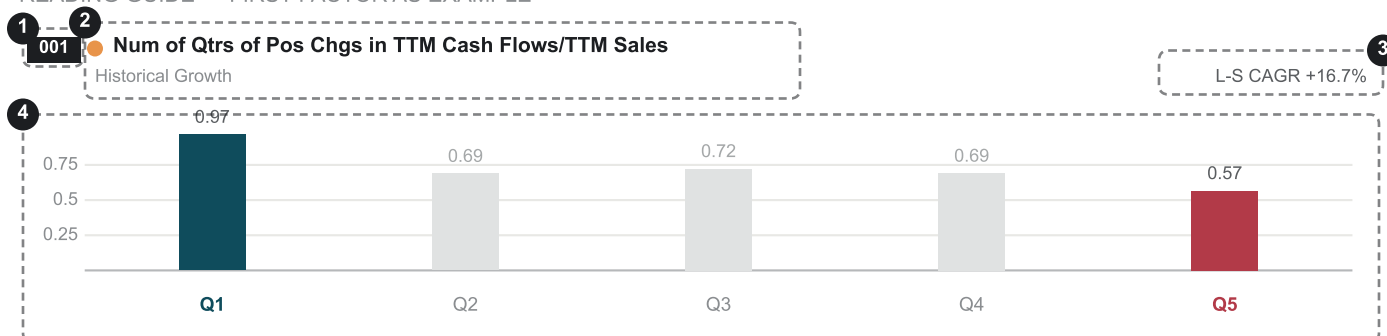
Quintiles are selected as long or short by performance against a spread-based threshold — not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

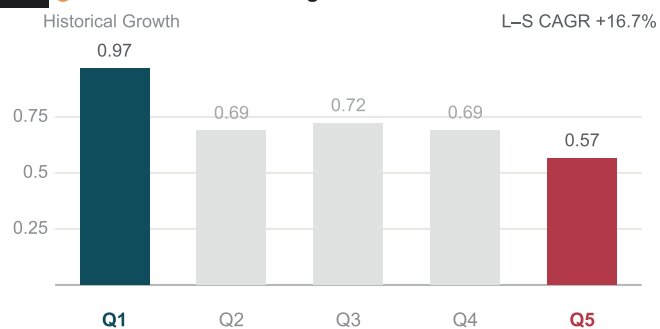
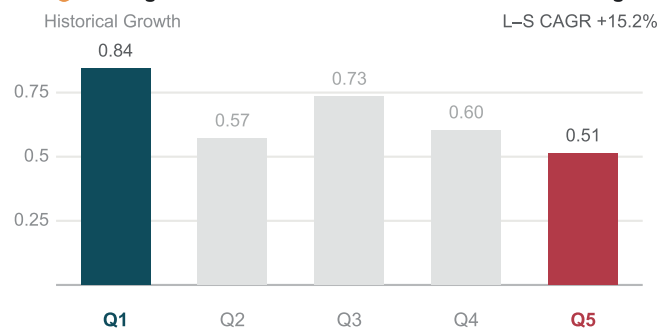
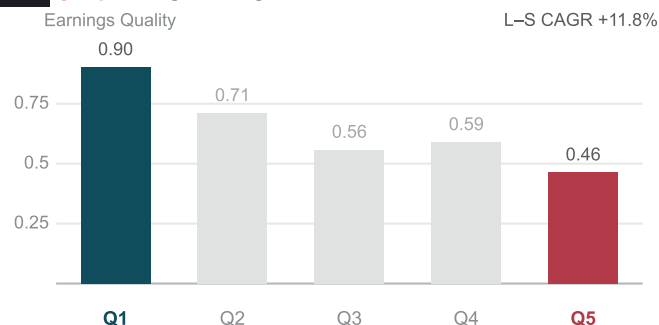
FACTOR STYLES

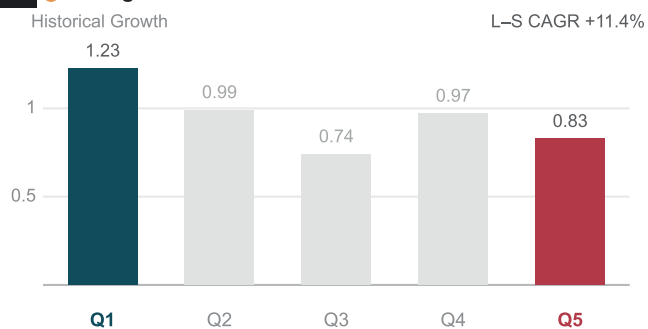
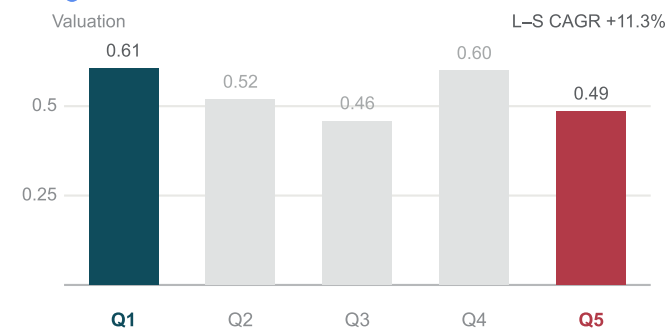
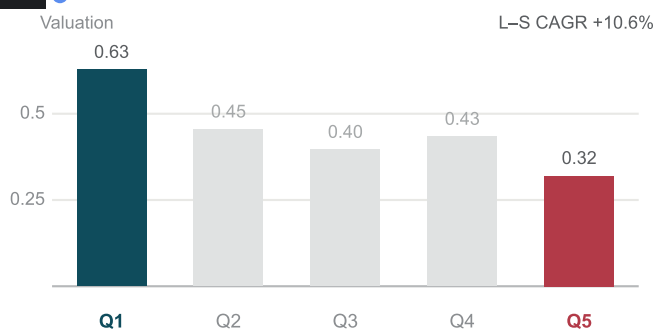
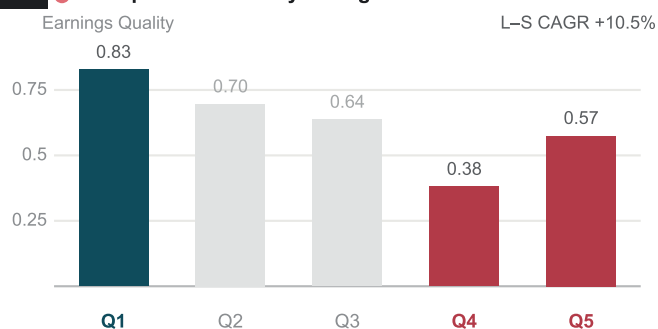
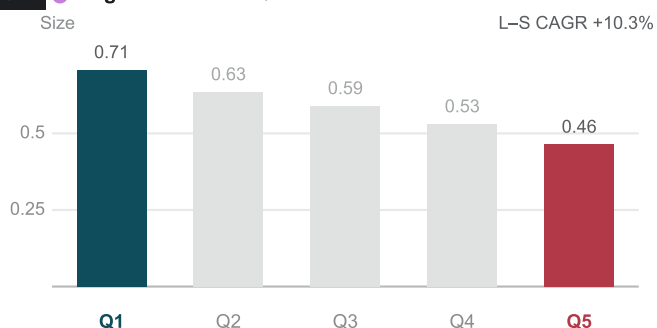
Valuation	41	Price Momentum	56
Earnings Quality	55	Size	1
Analyst Expectations	24	Historical Growth	55
Capital Efficiency	28	Volatility	10

001 In current model portfolio (50 factors)

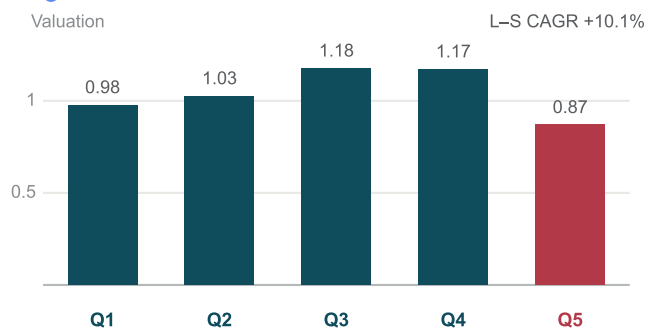
READING GUIDE — FIRST FACTOR AS EXAMPLE



001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-M...****003 ● Pretax Income Per Share to Price****004 ● Current Liability to Price****005 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flo****006 ● Operating Earnings to Assets Ratio**

007 ● Change of Asset Turnover**008 ● Assets to Price Ratio****009 ● Sales to Price Ratio****010 ● Unexpected Inventory Change****011 ● Log of Total Last Quarter Assets****012 ● Inverse P/E Ratio Adj for Growth and Yield**

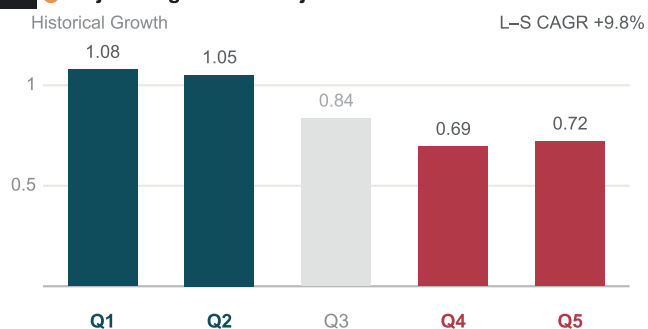
013 ● Indicated Annual Dividends to Price Ratio



014 ● Cash to Price



015 ● Adj 1Y Chg in Asset Adjusted FCF



016 ● Long Term Debt to Equity Ratio

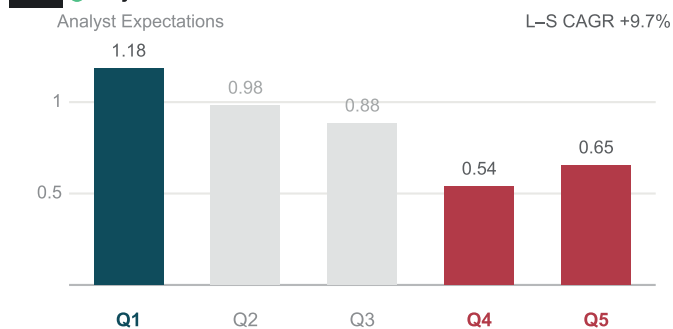
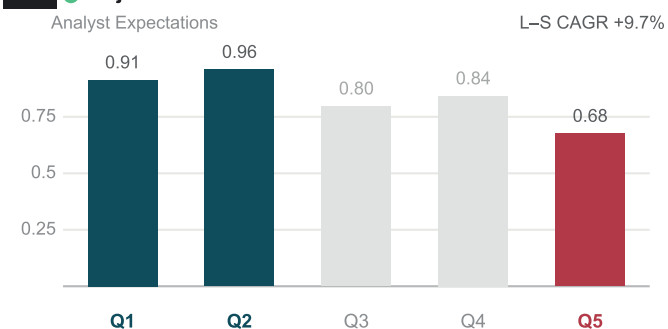
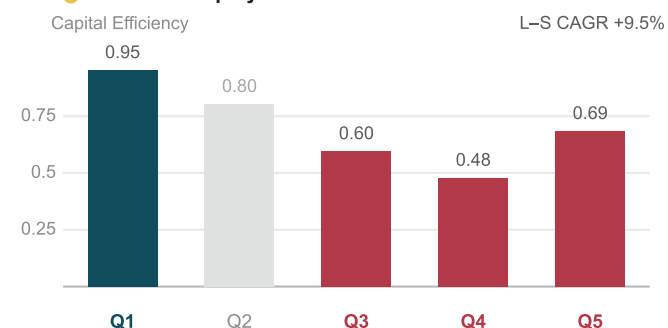
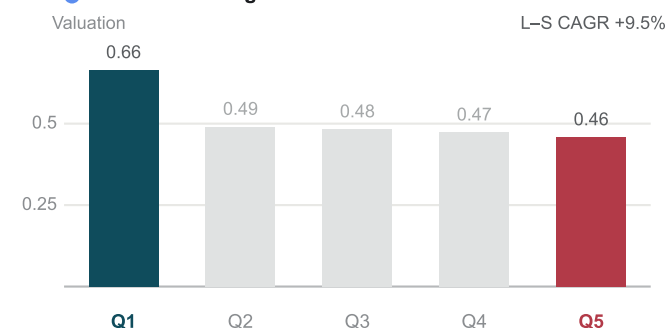


017 ● Return on Assets

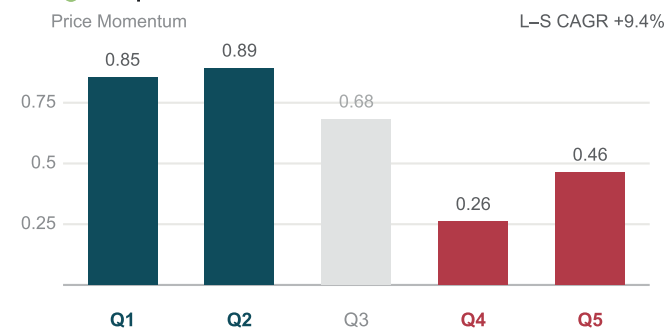


018 ● Number of Consecutive Qtrs of Positive Changes in TTM Quart...



019 ● Gross Profit Margin**020 ● Adjusted 3M Revision in FY1 EPS Estimate****021 ● Adjusted Inverse PEG****022 ● Analyst Dispersion for FY2 EPS****023 ● Return on Equity****024 ● Forward Earnings to Price**

025 ● Sharpe Ratio



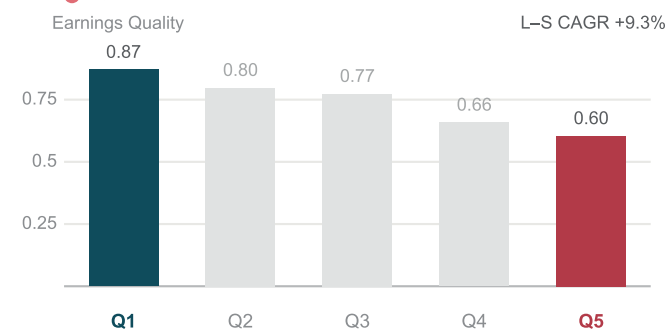
026 ● 12M Realized Price Volatility



027 ● Return on Invested Capital



028 ● Quick Ratio

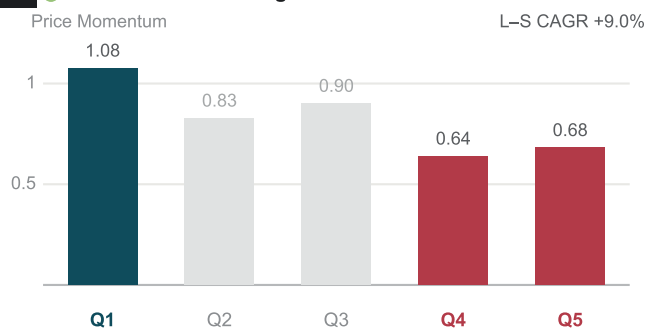
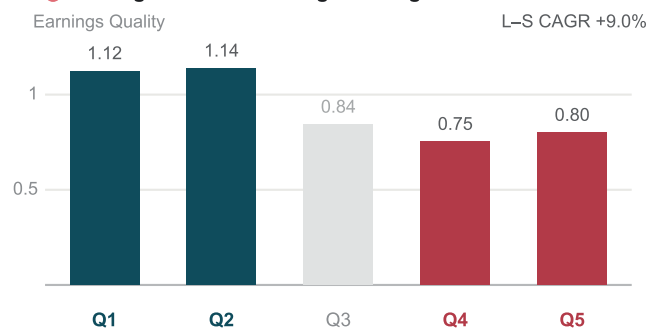
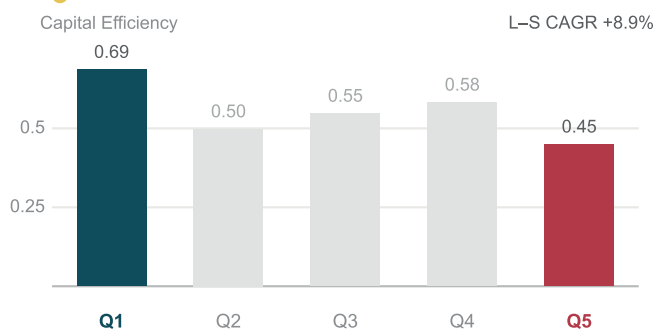
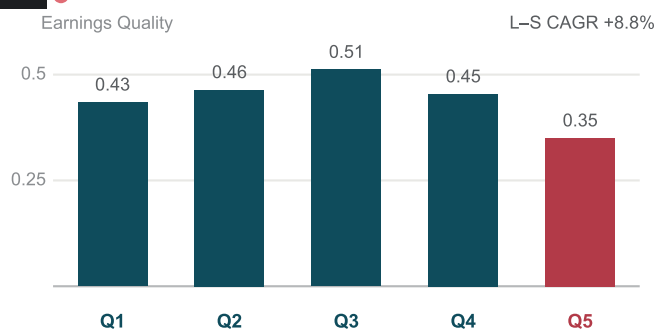
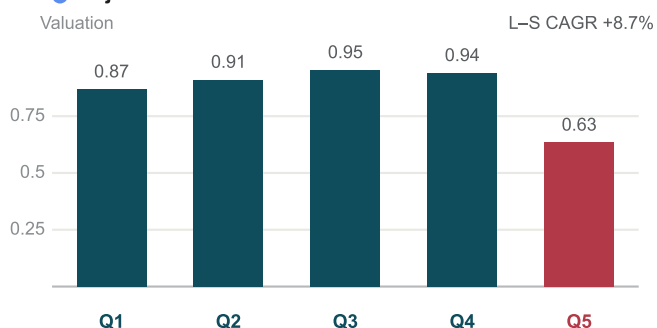
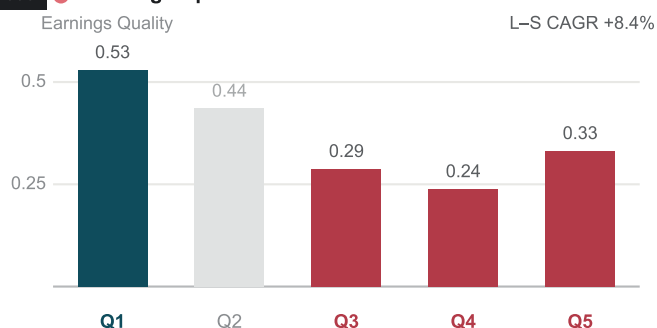


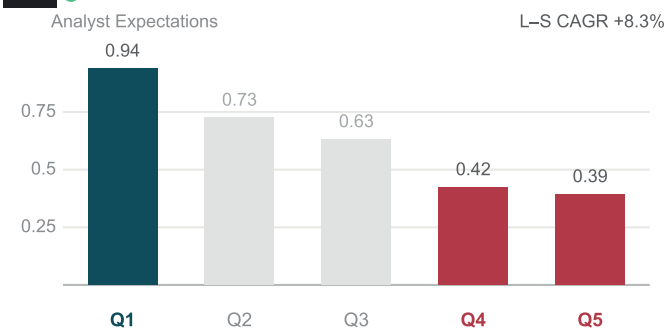
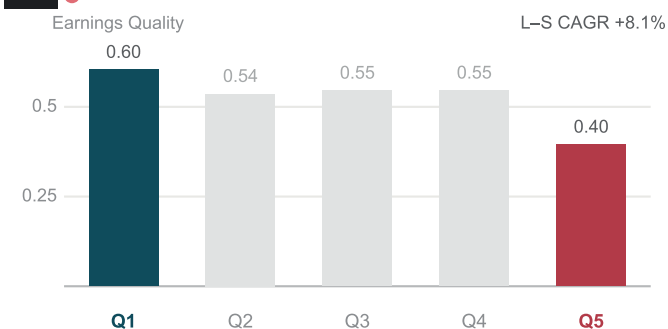
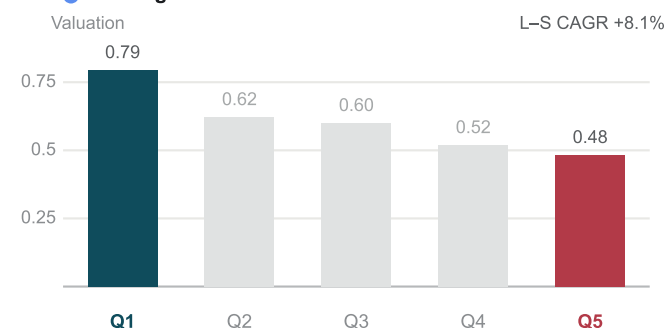
029 ● 3Y Chg in Operating Cash Flow to Price



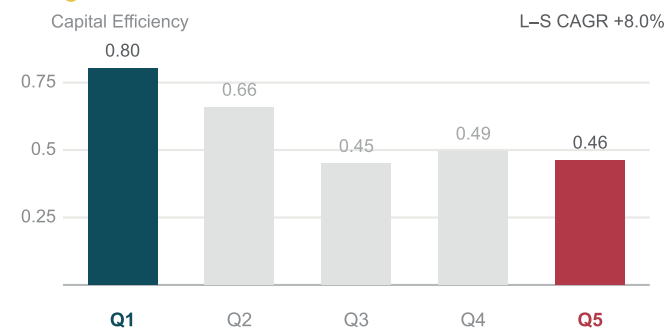
030 ● Operating Earnings to Price Ratio



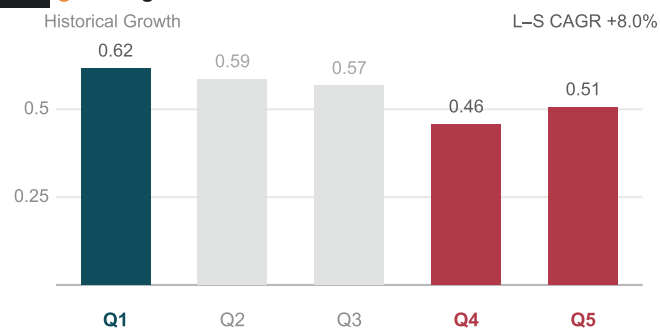
031 ● Short Interest Trading Volume**032 ● Trailing 12-Month Earnings/Trailing 12-Month Sales****033 ● Tobin's Q Ratio****034 ● ROA 20 Qtr Standard Deviation****035 ● Adj Growth Flow to Price****036 ● Working Capital to Sales**

037 ● 3M Revision in FY1 EPS Estimate**038 ● Adjusted Forward Earnings to Price****039 ● Normalized EP****040 ● Balance Sheet Accruals****041 ● Earnings to Price****042 ● Slope of 52W Price Trend**

043 ● EBITDA to Assets



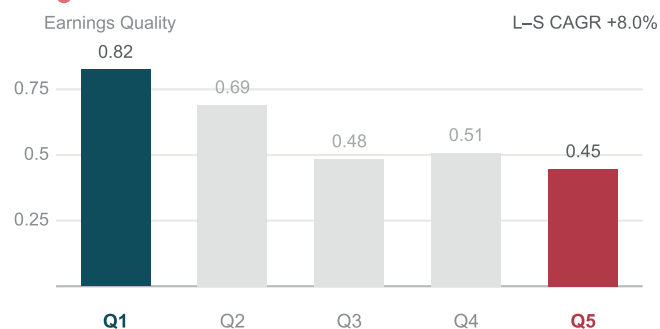
044 ● 3Y Chg in Cash Flow to Price



045 ● Consecutive Quarters of Declines in TTM Overhead/Sales



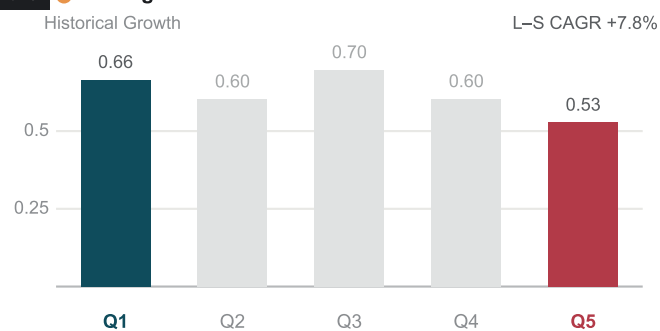
046 ● Cash Flow to Total Asset



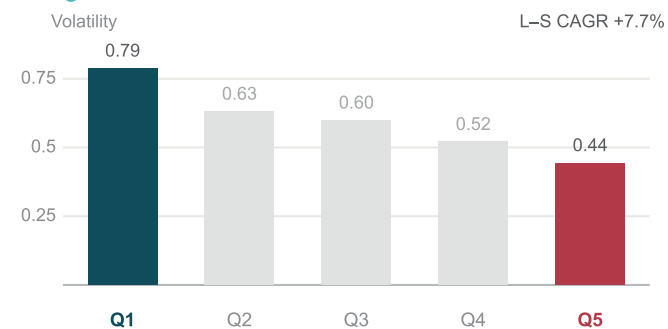
047 ● Book Leverage



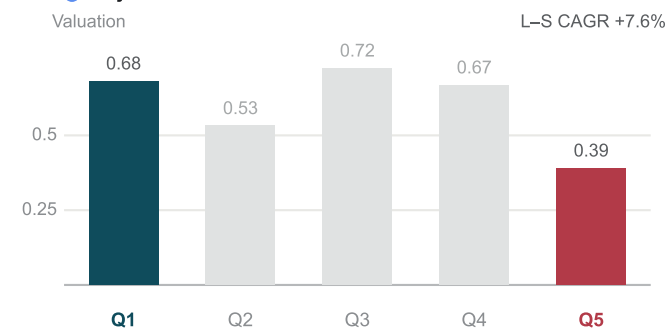
048 ● 1Y Chg in Sales to Price



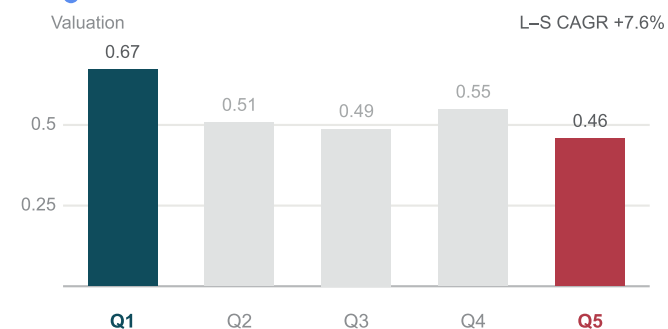
049 ● 60M CAPM Beta



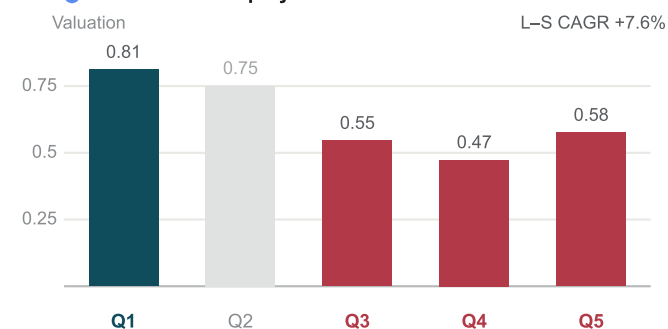
050 ● Adj Free Cash Flow to Price



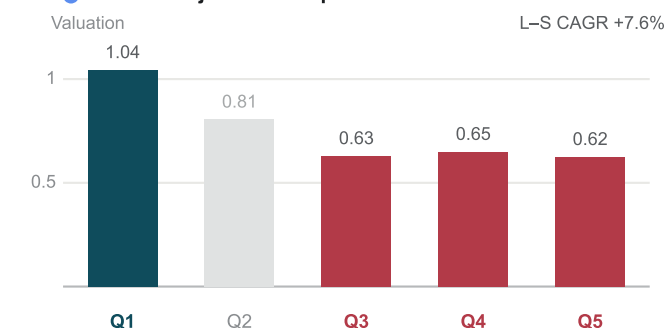
051 ● Sales to EV Ratio



052 ● Cash Flow to Equity

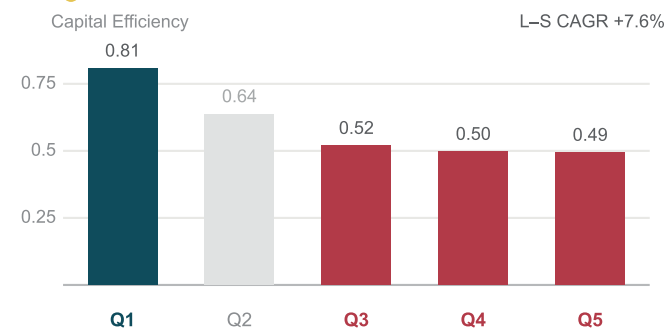
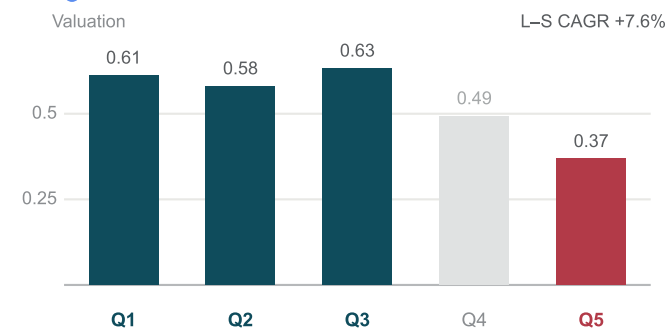
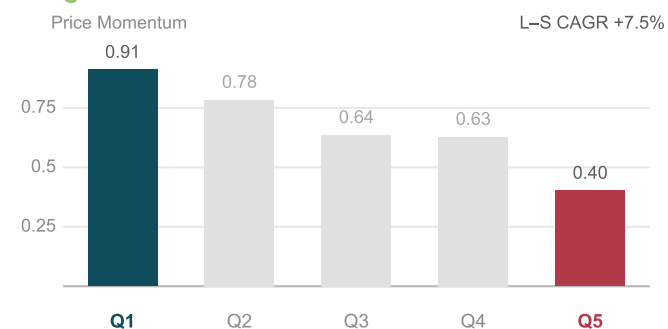
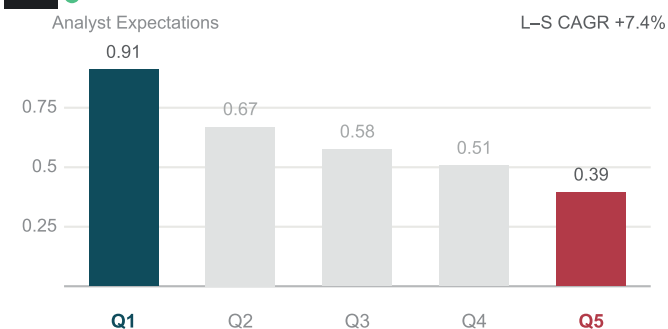
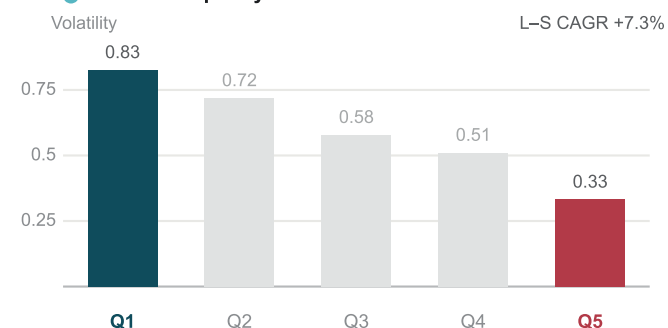
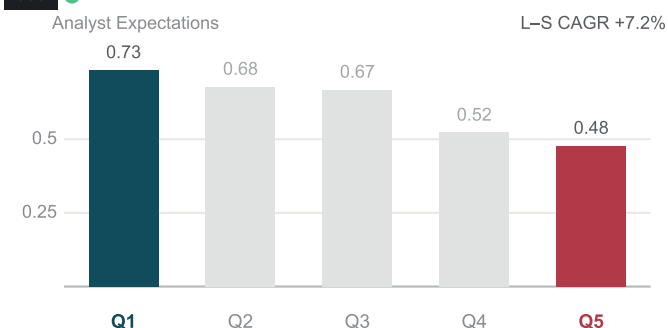


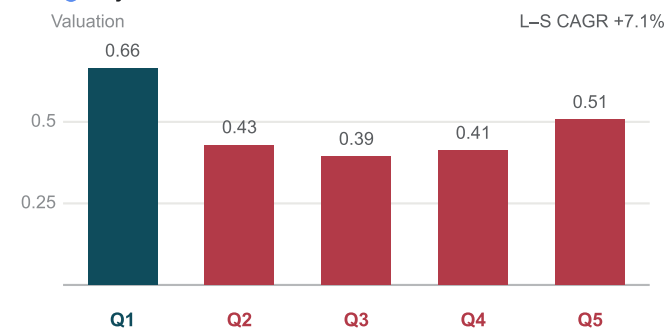
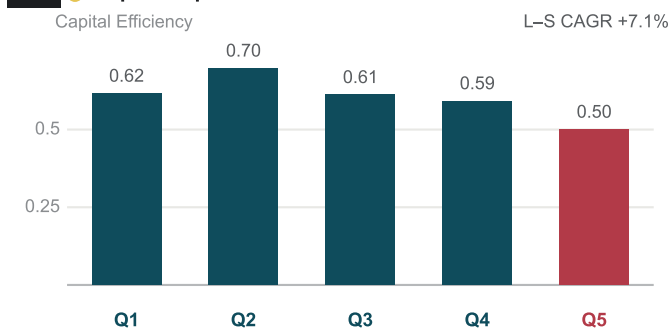
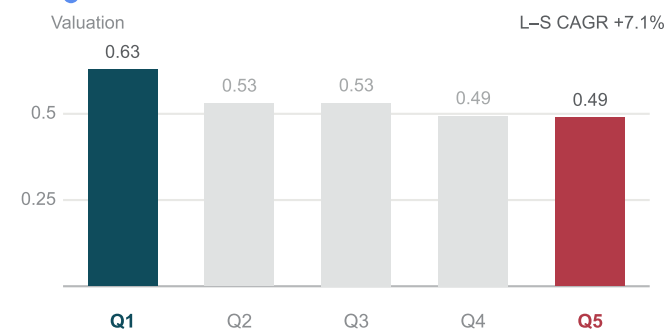
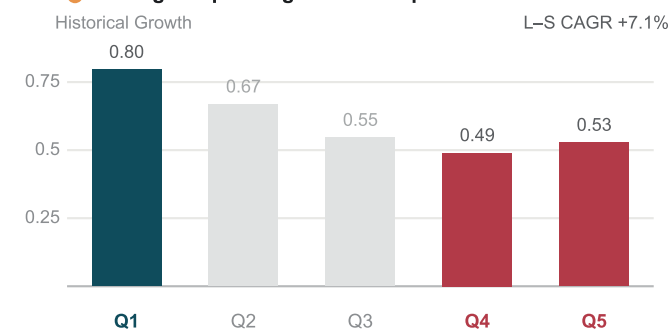
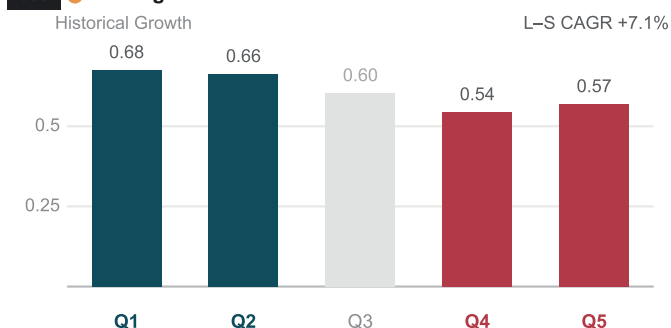
053 ● Cash to Adjusted Enterprise Value



054 ● Expected LTG



055 ● **EBIT to Assets**056 ● **Dividends to Price Ratio**057 ● **Short Interest Ratio**058 ● **3M Revision in FY2 EPS Estimate**059 ● **Amihud Illiquidity Measure**060 ● **Inverse PEG**

061 ● **Adjusted EBIT to Price**062 ● **Capital Expenditure to Total Assets**063 ● **Net Cash Flow to Price**064 ● **3Y Chg in Operating Cash Flow per Share**065 ● **1Y Chg in Operating Margin**066 ● **1Y Chg in Cash Flow to Price**

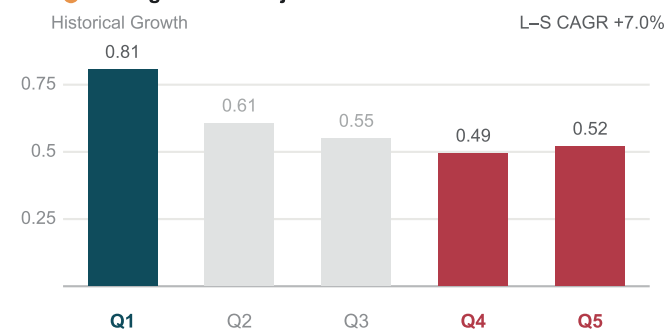
067 ● 50 Day to 200 Day Stock Price



068 ● Adjusted Book to Enterprise Value



069 ● 1Y Chg in Asset Adjusted EPS



070 ● Earnings Surprise

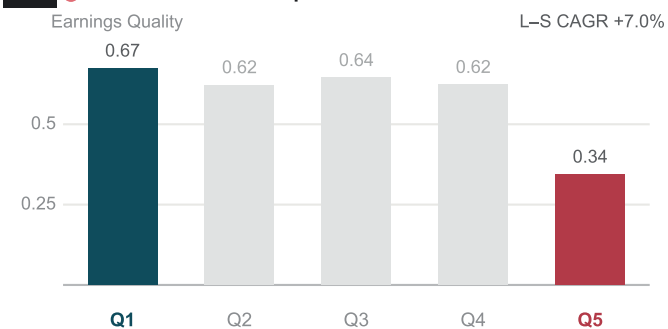
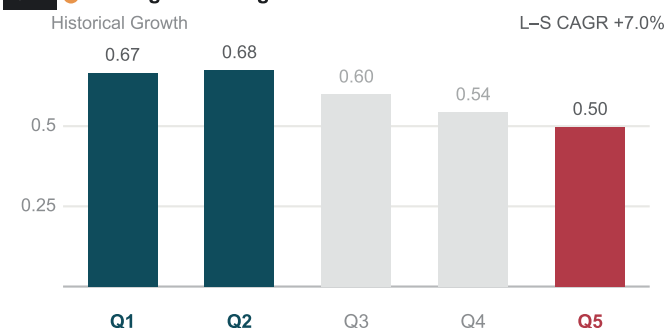


071 ● Street Revision Magnitude



072 ● Cash Flow to Invested Capital

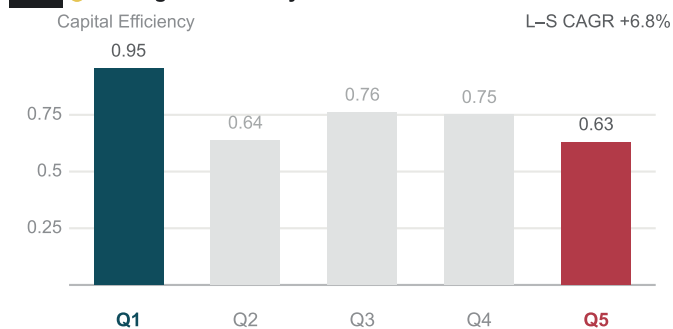


073 ● Sales to Invested Capital**074 ● 39W Lagged Return****075 ● Operating Cash Flow to Invested Capital****076 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow****077 ● 1Y Chg in Earnings to Price****078 ● Operating Cash Flow to Asset**

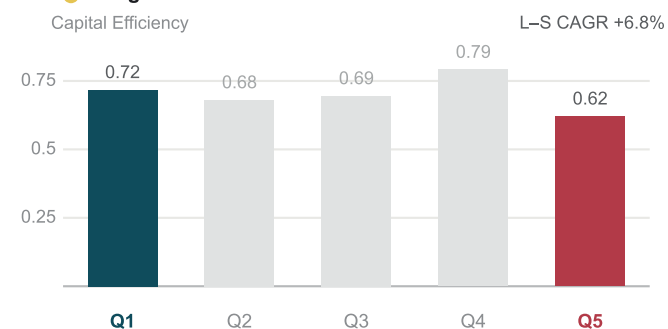
079 ● EBITDA to Enterprise Value



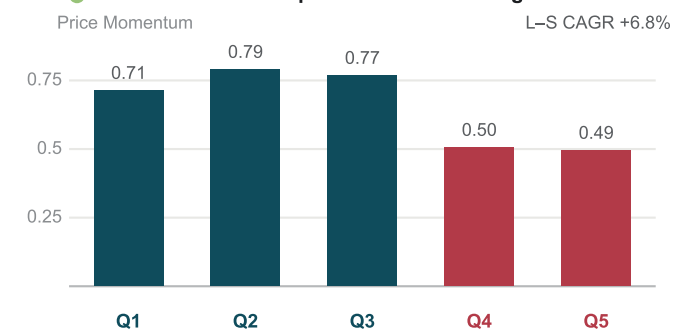
080 ● 1Y Chg in Stock Buybacks



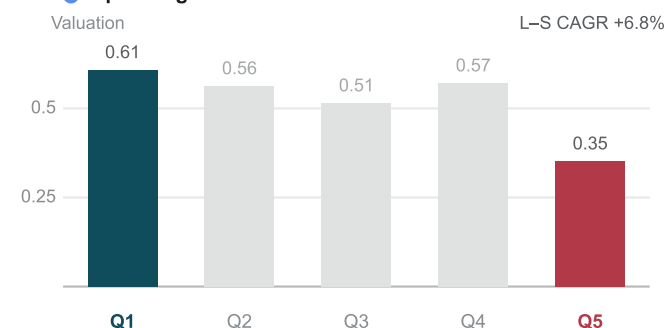
081 ● Long Term Debt to Assets Ratio



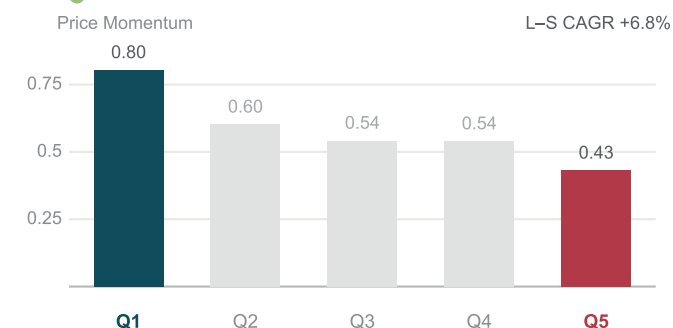
082 ● Institution Ownership Level - Active Manager



083 ● Operating Cash Flow to Price



084 ● 15/36 Week Price Ratio



085 ● 26W Relative Price Strength



086 ● Adj 1Y Chg in Earnings per Share



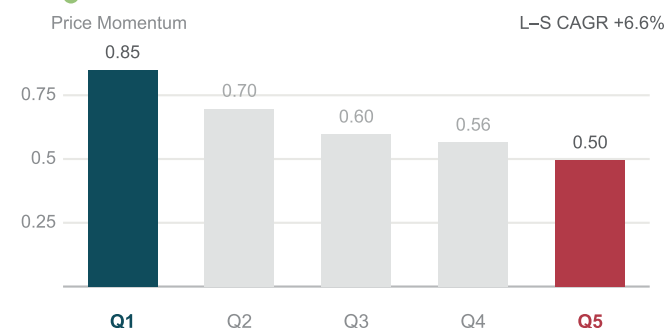
087 ● 90 Day Coefficient of Variation



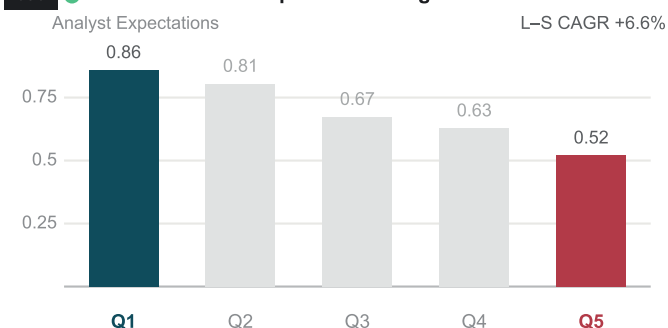
088 ● Solvency Ratio

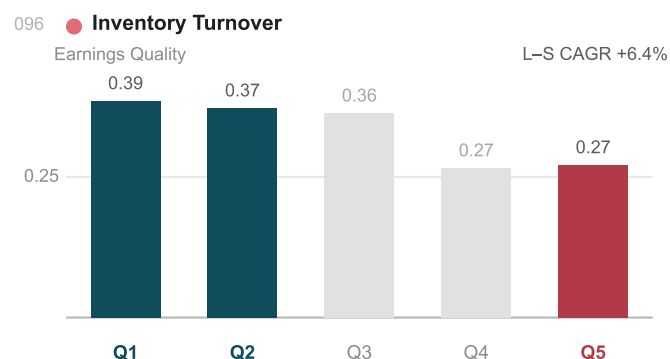
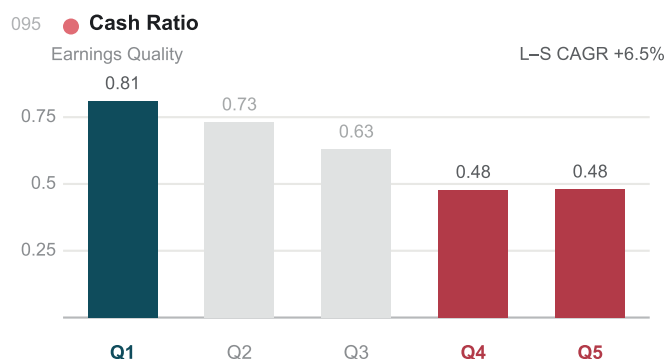
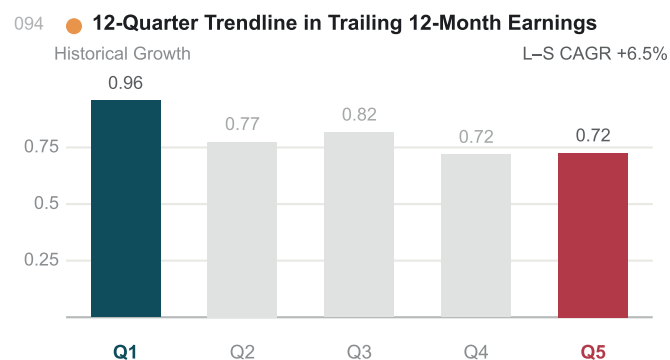
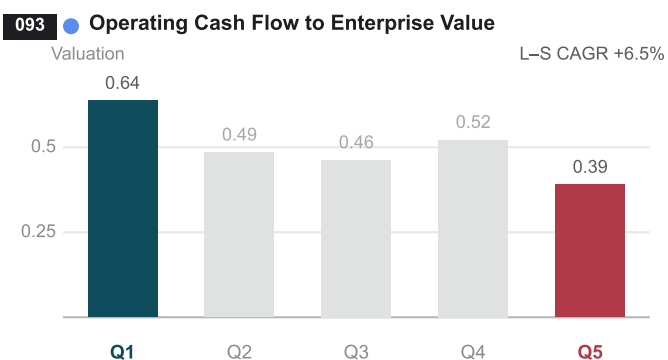
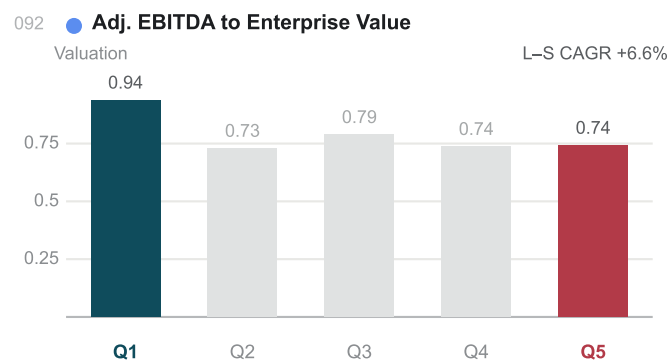
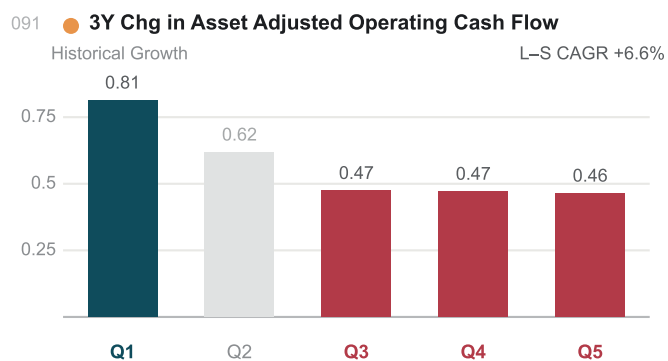


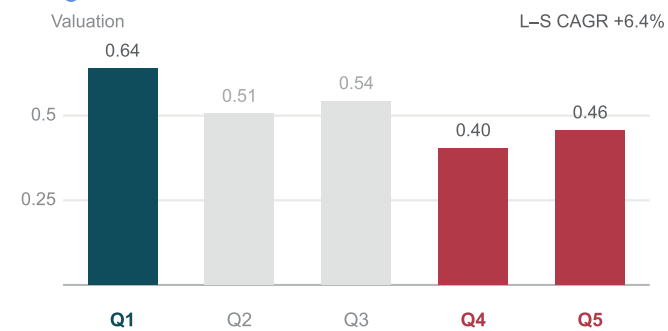
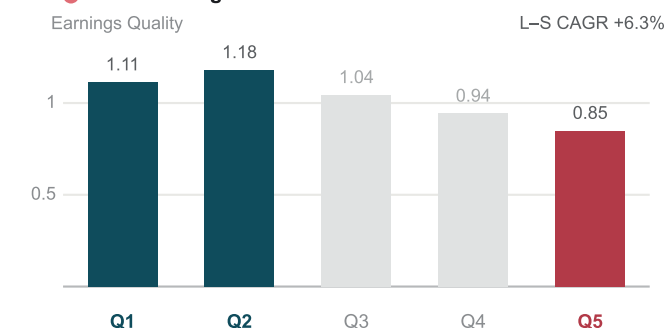
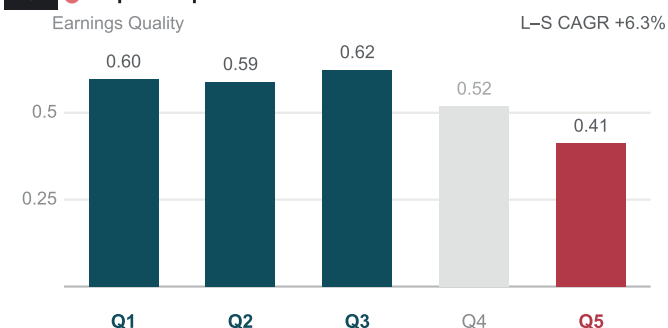
089 ● 9 Month Price Momentum

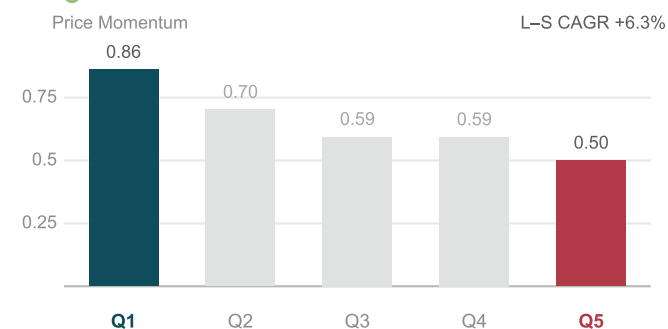
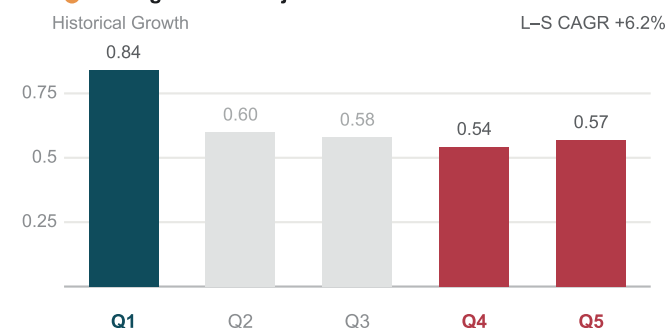


090 ● Standardized Unexpected Earnings





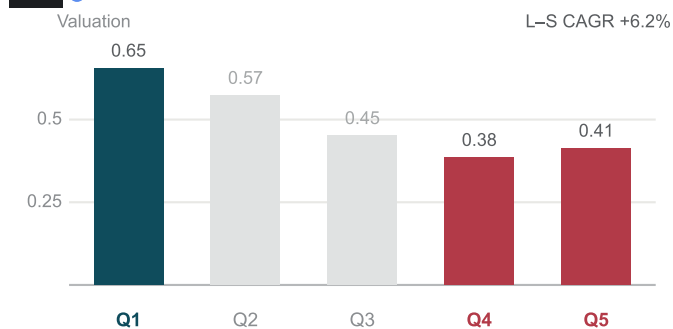
097 ● **Book to Price**098 ● **6M Chg in Target Price**099 ● **Adjusted Revision Magnitude**100 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**101 ● **EBITDA Margin**102 ● **Capital Expenditure to Sales**

103 ● Free Cash Flow to Enterprise Value**104 ● 1M Realized Price Volatility****105 ● EBITDA to Price****106 ● 4W to 52W Price Oscillator****107 ● Percent Above 260-Day Low****108 ● 1Y Chg in Asset Adjusted Cash Flow**

109 ● Growth Flow to Price



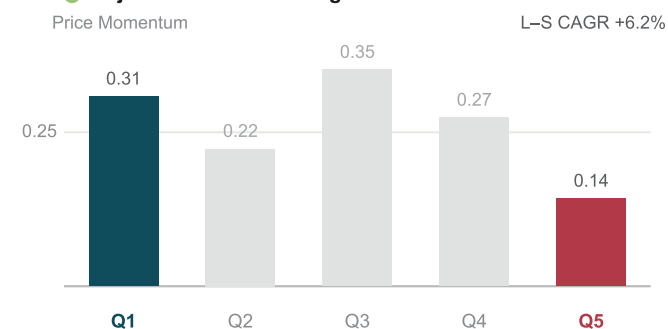
110 ● Cash to EV Ratio



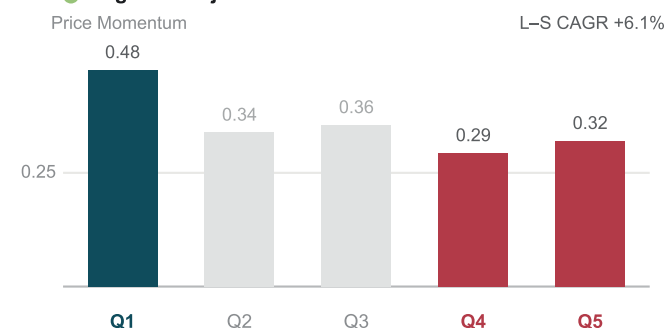
111 ● Year over Year Change in SGA to Sales



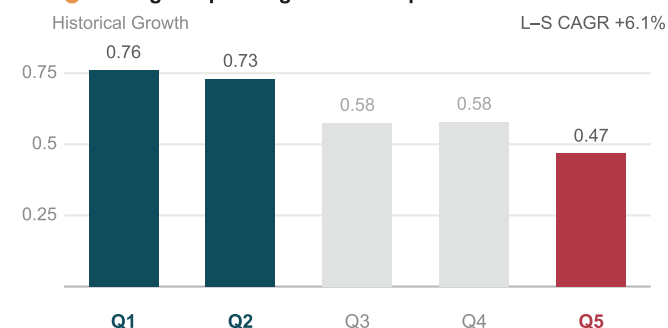
112 ● Adjusted 6 Month Average Share Turn Over



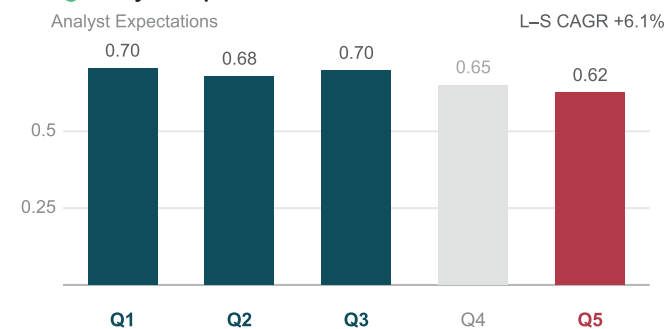
113 ● Log of Unadjusted Stock Price



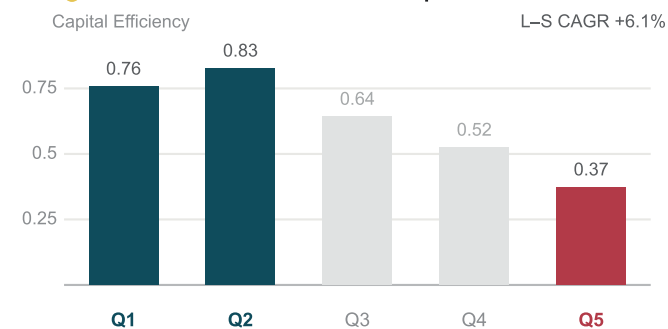
114 ● 1Y Chg in Operating Cash Flow per Share



115 ● Analyst Dispersion for FY1 EPS



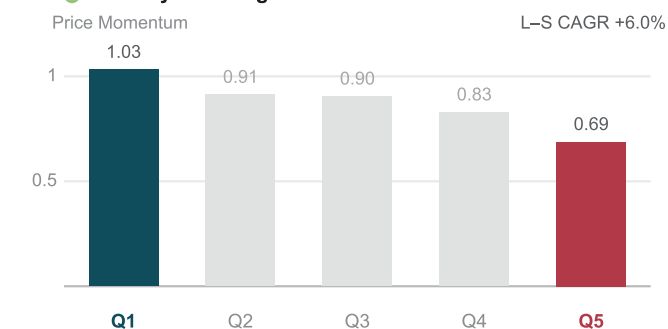
116 ● Cash Flow Return on Invested Capital



117 ● 1Y Chg in Asset Adjusted Free Cash Flow



118 ● Stability of Change in Number of Holders

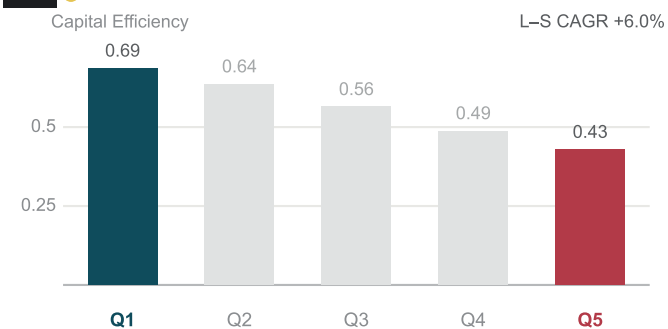
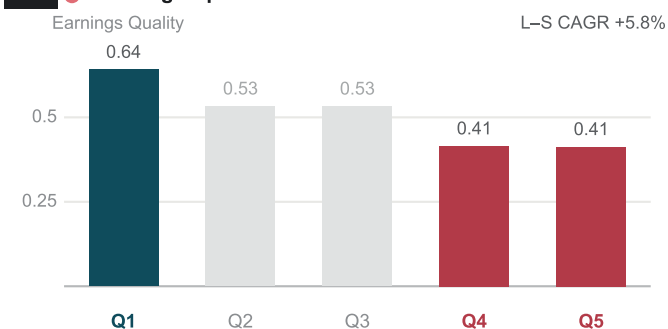
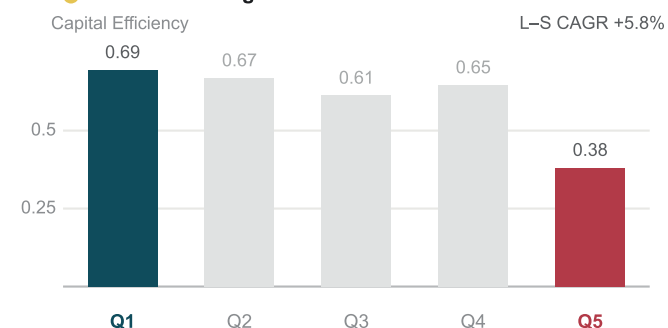


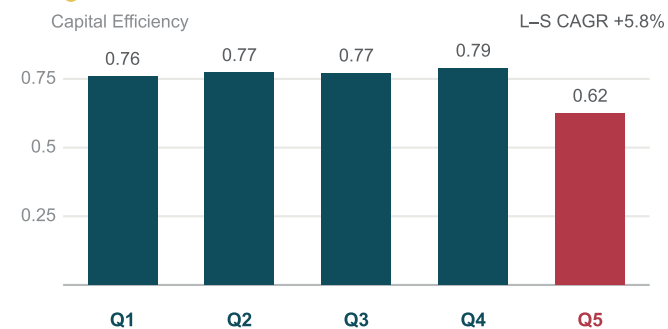
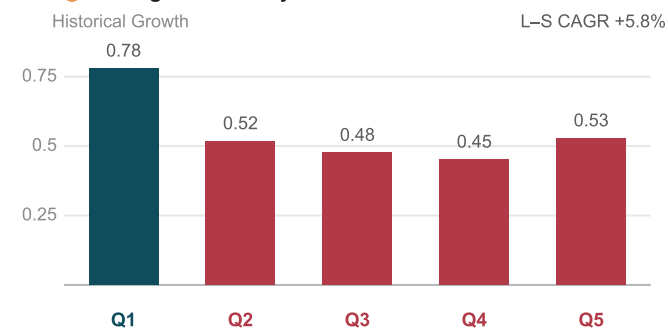
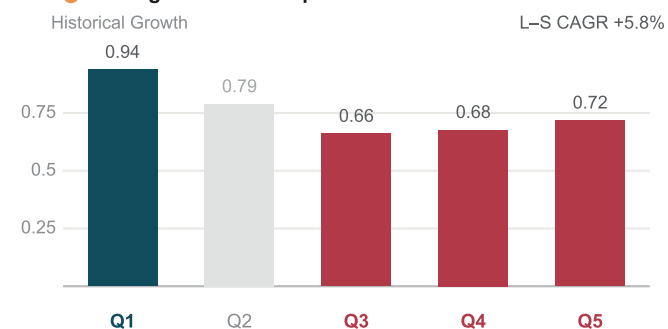
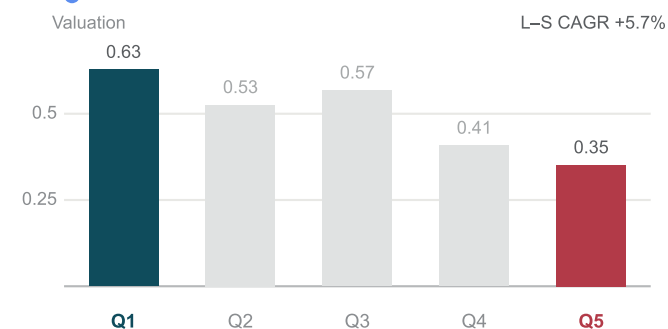
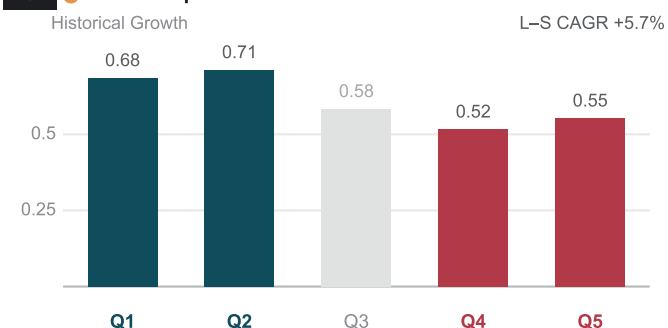
119 ● Operating Cash Flow Ratio

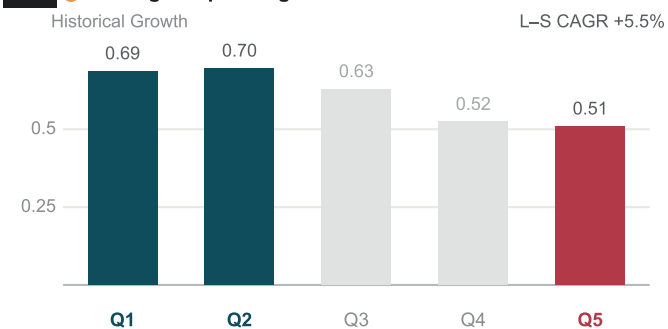
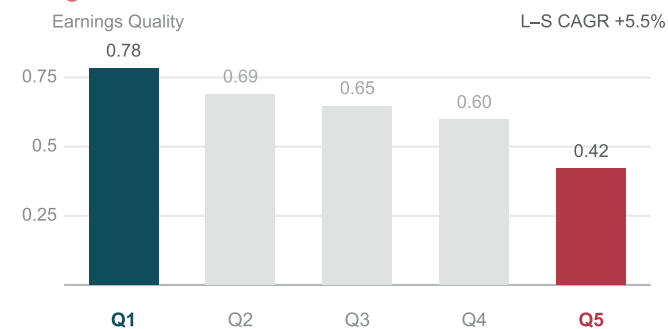


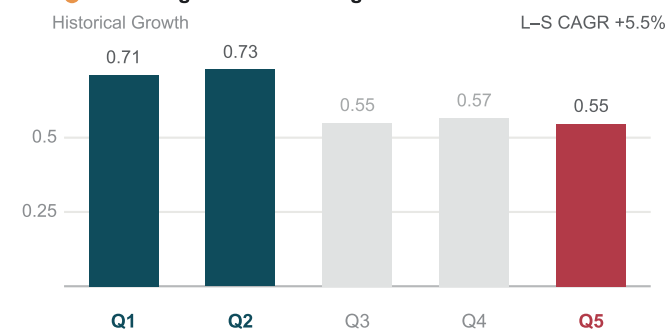
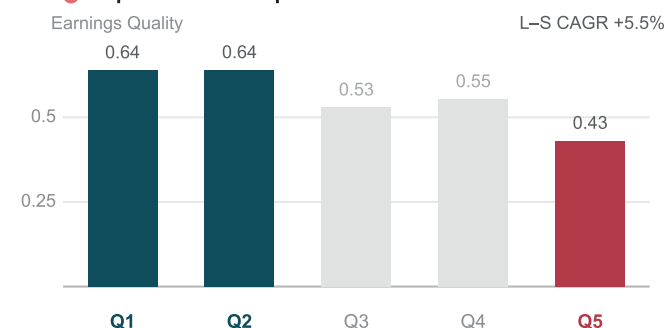
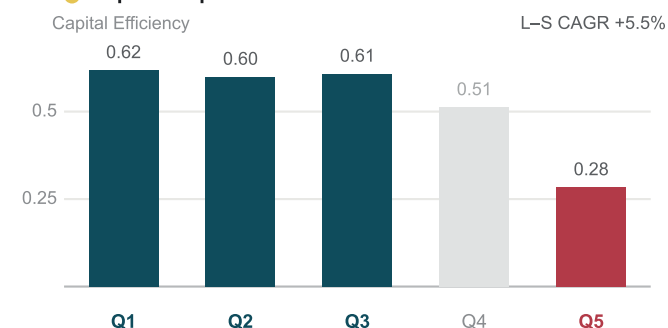
120 ● Equity Turnover



121 ● Receivables Turnover Ratio**122 ● Adjusted Interest Coverage Ratio****123 ● 3Y Chg in Free Cash Flow to Price****124 ● Working Capital Accruals****125 ● Interest Coverage Ratio****126 ● Forward Free Cash Flow to Price Ratio**

127 ● **Debt to Assets Ratio**128 ● **3Y Chg in Asset Adjusted Cash Flow**129 ● **3Y Chg in Cash Flow per Share**130 ● **Free Cash Flow to Price**131 ● **Dividend per Share Growth**132 ● **12-Quarter Trendline in Trailing 12-Month Cash Flows**

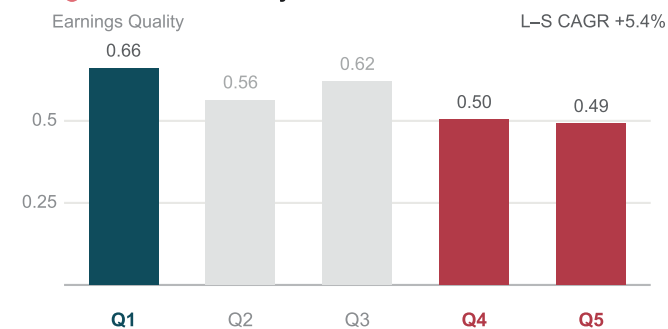
133 ● 1Y Chg in Operating Cash Flow to Price

134 ● Cash Burn Rate

135 ● 52-Week Volume Price Trend with 20-Day Lag

136 ● 3Y Average Annual Earnings Growth

137 ● Depreciation to Capex Ratio

138 ● Capital Acquisition Ratio


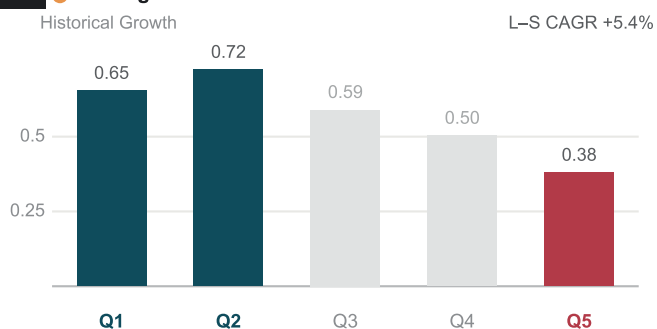
139 ● Free Cash Flow Stability



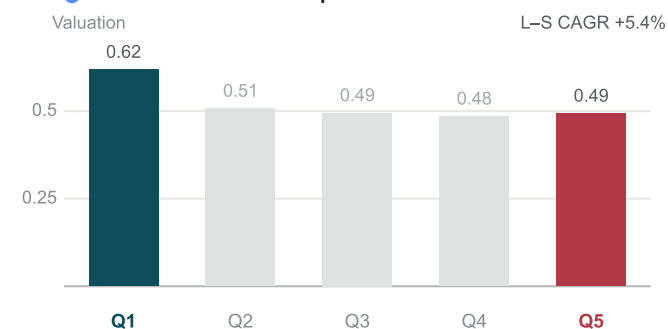
140 ● Net Income Stability



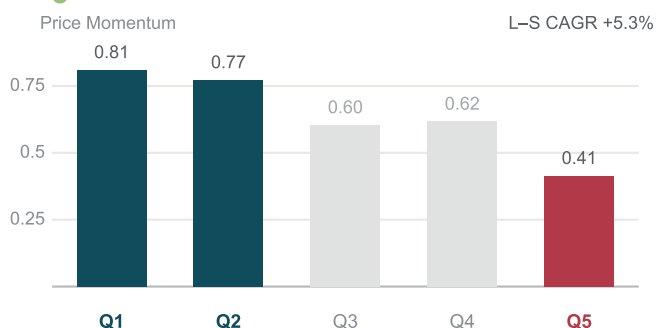
141 ● 1Y Chg in Free Cash Flow to Price



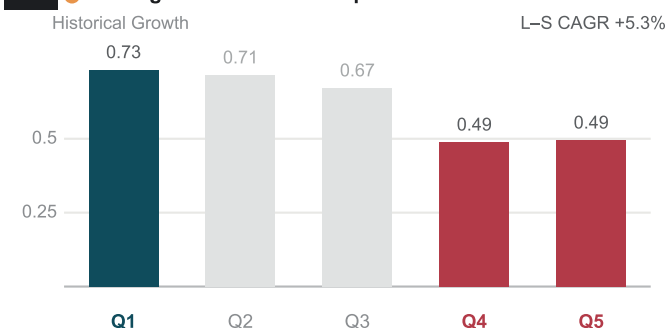
142 ● Net Cash Flow to Enterprise Value

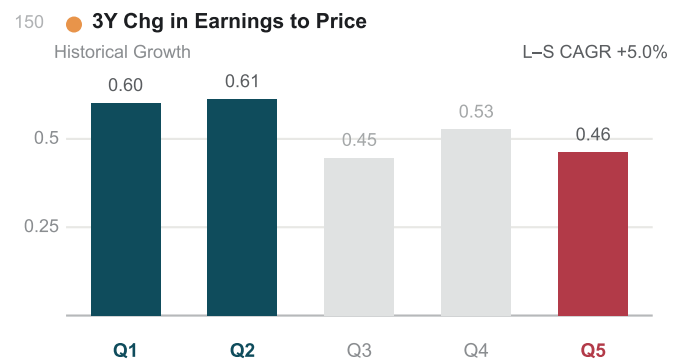
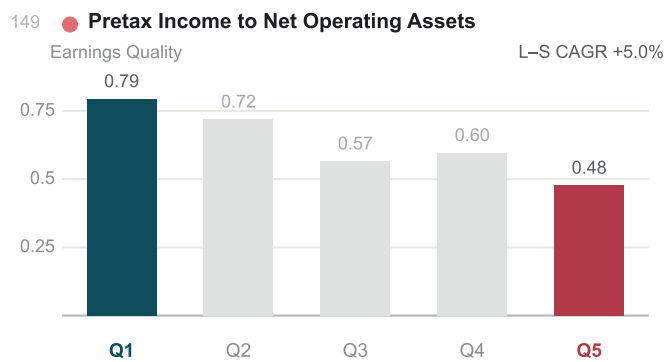
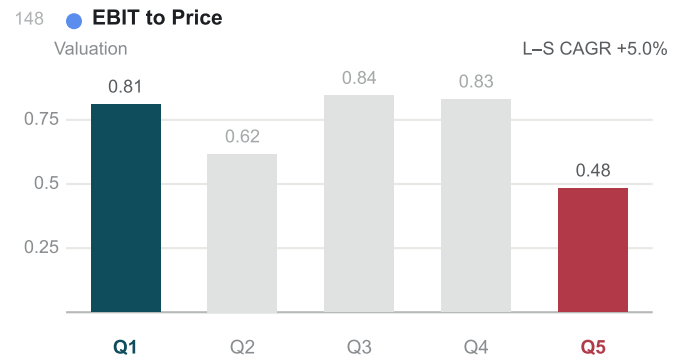
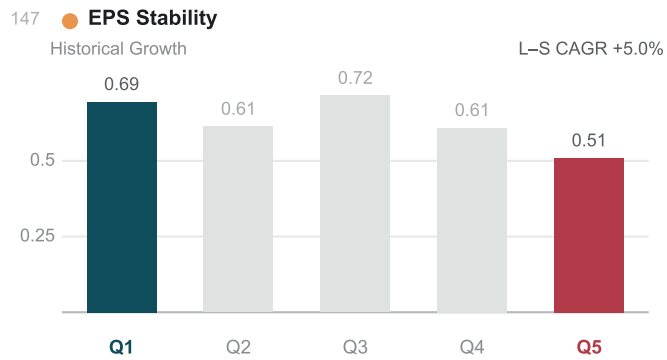
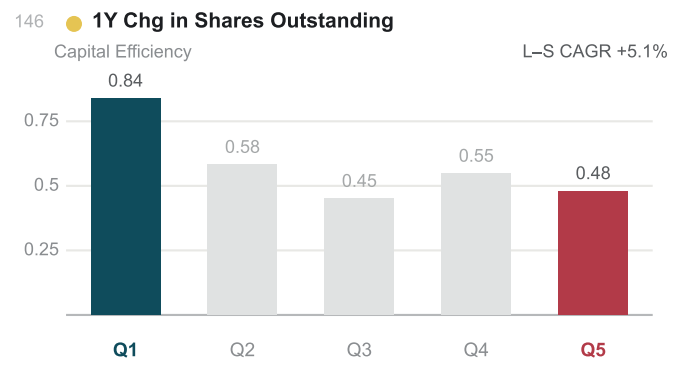
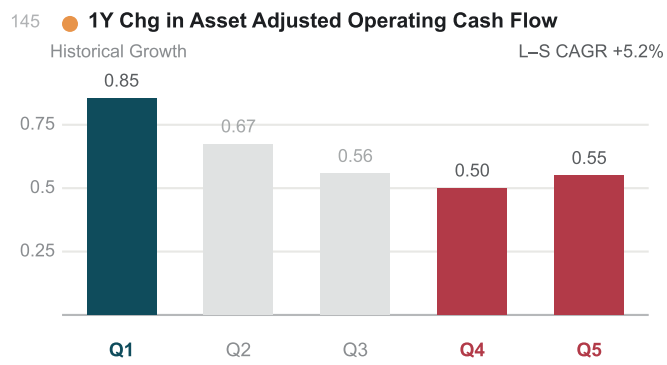


143 ● 12M - 1M Price Momentum

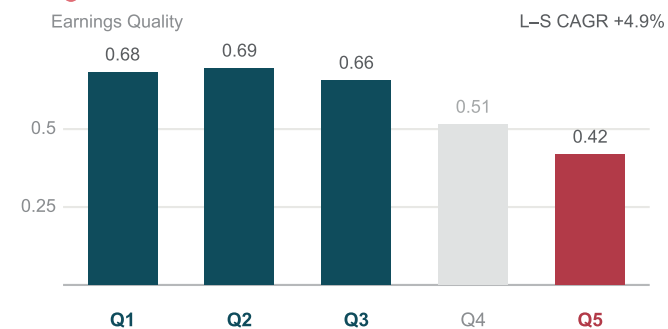


144 ● 1Y Chg in Free Cash Flow per Share

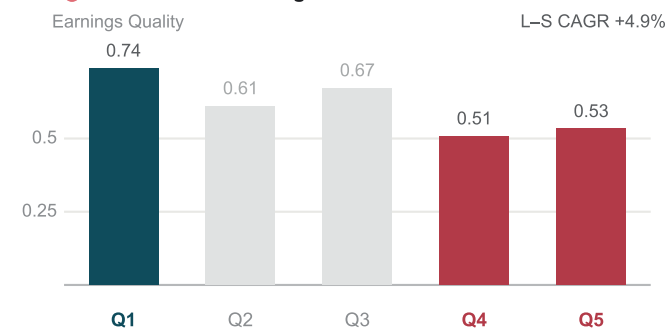




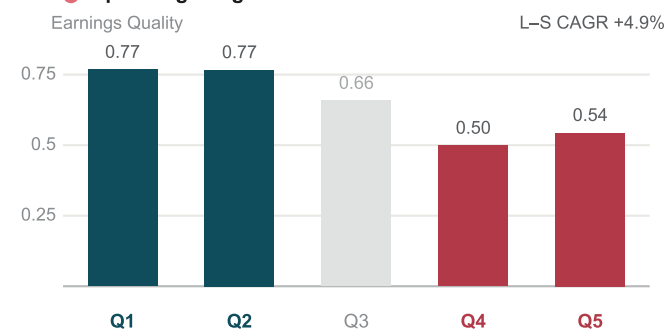
151 ● Free Cash Flow to Sales



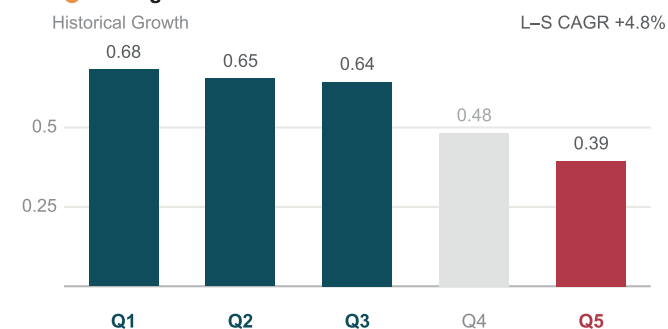
152 ● Year over Year Change of EPS to Sales



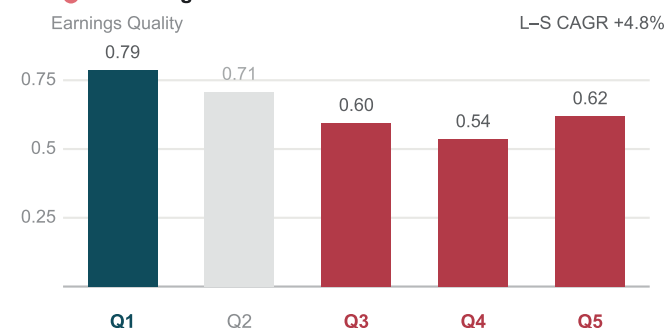
153 ● Operating Margin



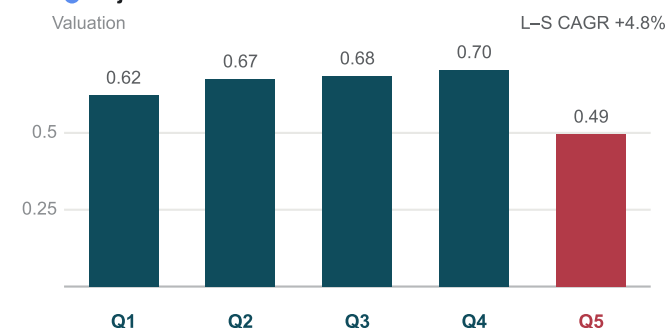
154 ● 3Y Chg in Sales to Price



155 ● EBIT Margin



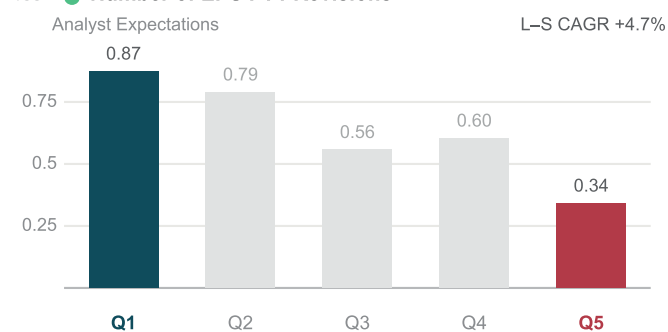
156 ● Adj Forward Free Cash Flow to Price Ratio



157 ● 3Y Chg in Free Cash Flow per Share



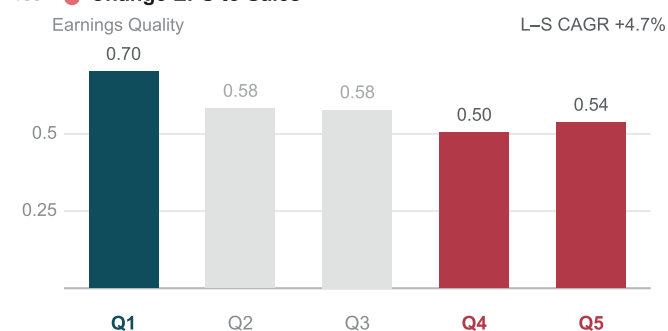
158 ● Number of EPS FY1 Revisions



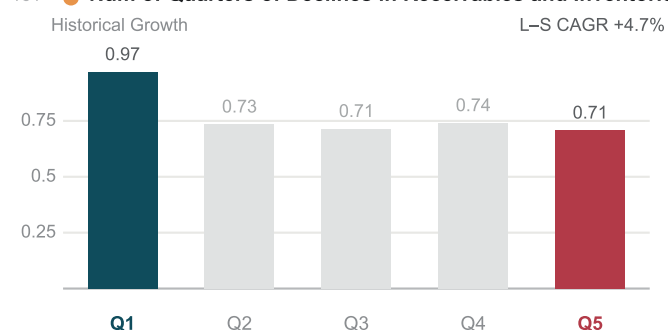
159 ● Working Capital to Assets Ratio



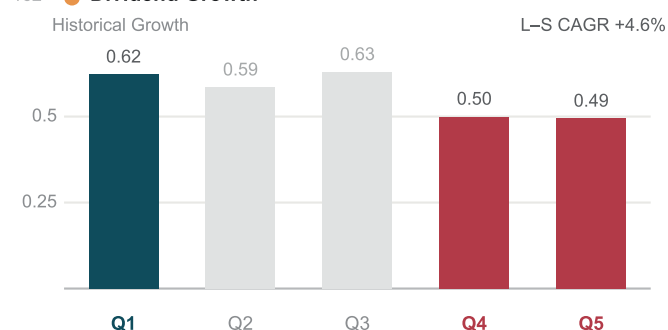
160 ● Change EPS to Sales

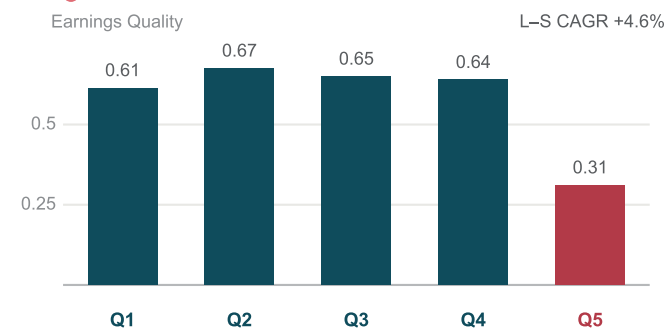
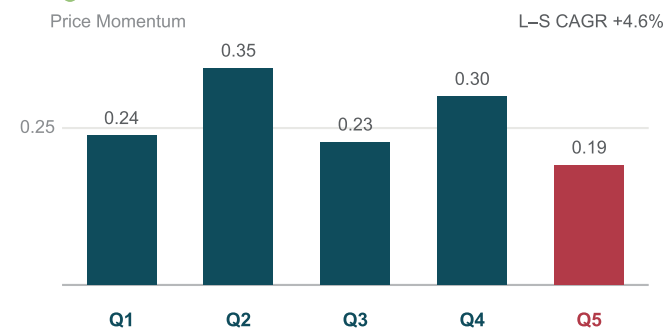
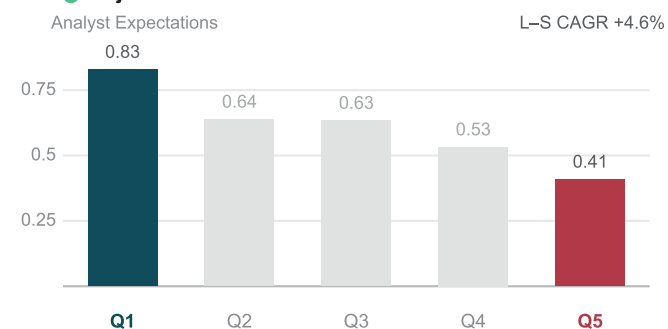
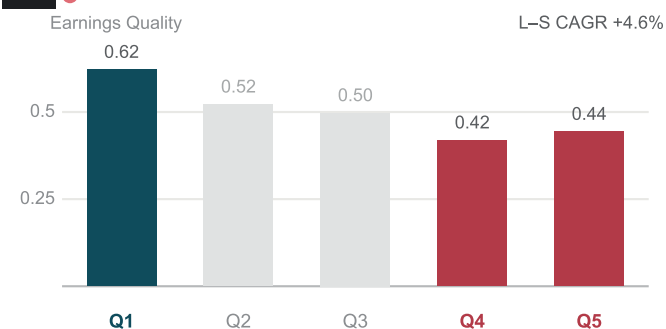
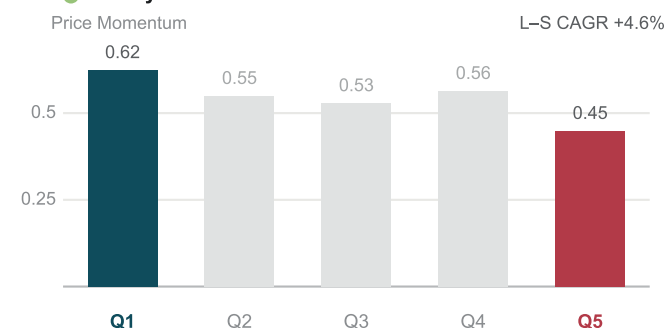
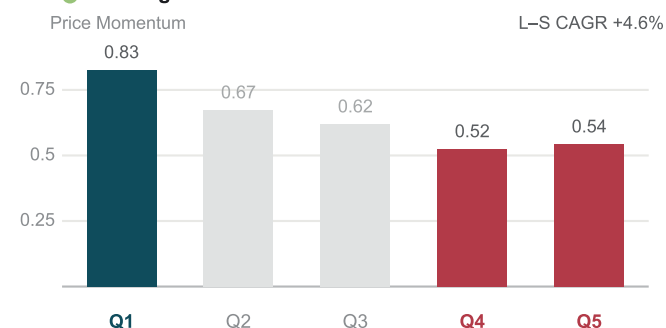


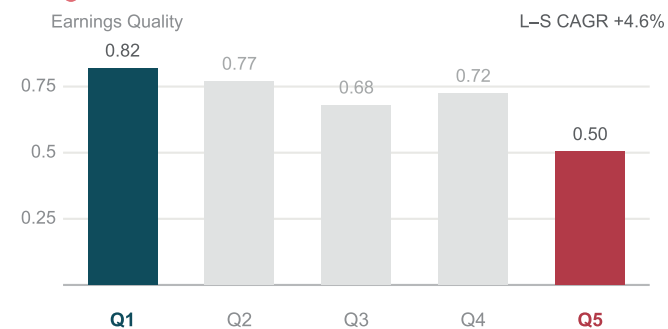
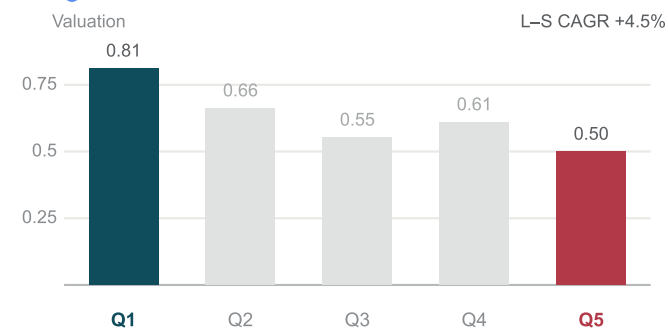
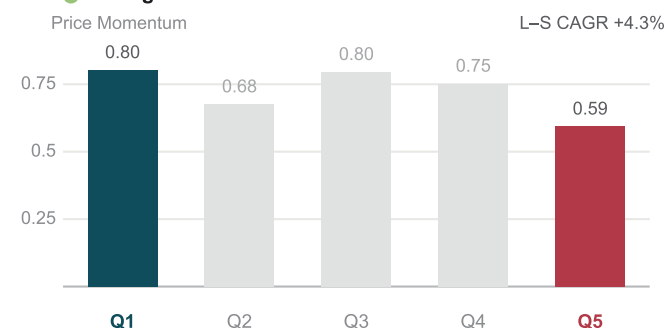
161 ● Num of Quarters of Declines in Receivables and Inventories/...

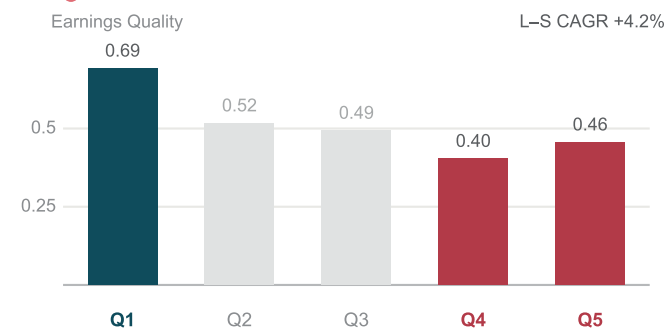
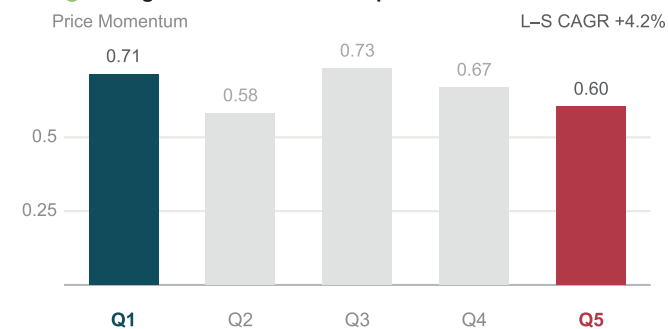
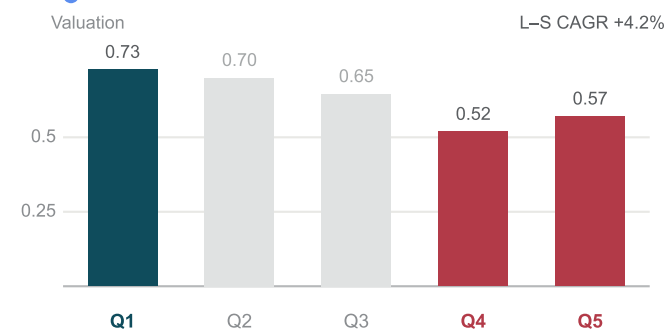
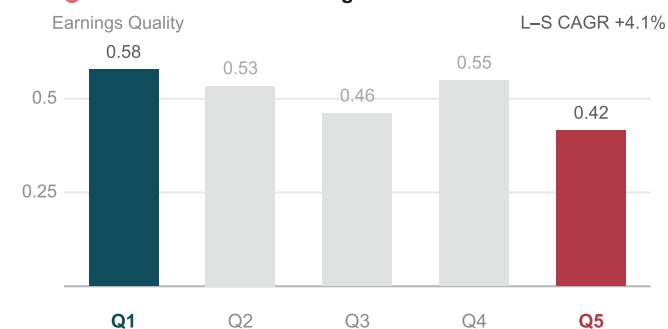
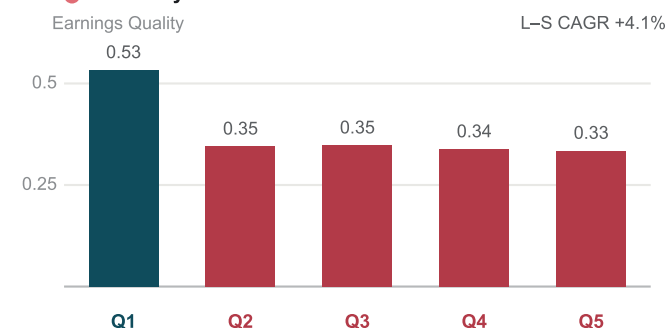


162 ● Dividend Growth

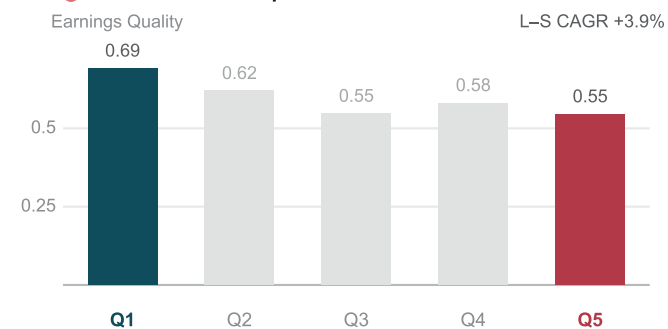


163 ● **Asset Turnover**164 ● **Share Turnover**165 ● **Adj Number of EPS FY2 Revisions**166 ● **Accrual Ratio - Cash Flows**167 ● **60 Day Coefficient of Variation of Volume to Price**168 ● **52W High Low**

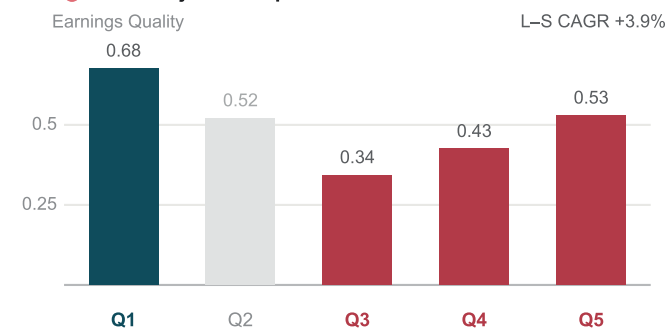
169 ● **Current Ratio**170 ● **Cash to Total Assets**171 ● **3Y Chg in Sales Turnover**172 ● **6M Momentum in Trailing 12M Sales**173 ● **Change in Number of Holders**174 ● **3Y Chg in Asset Adjusted Free Cash Flow**

175 ● **SG&A to Sales**176 ● **Foreign Institution Ownership**177 ● **Net Current Assets to Price Ratio**178 ● **Sales to Gross Profit Margin Growth**179 ● **Risk Adjusted Relative Strength**180 ● **Inventory to Assets Ratio**

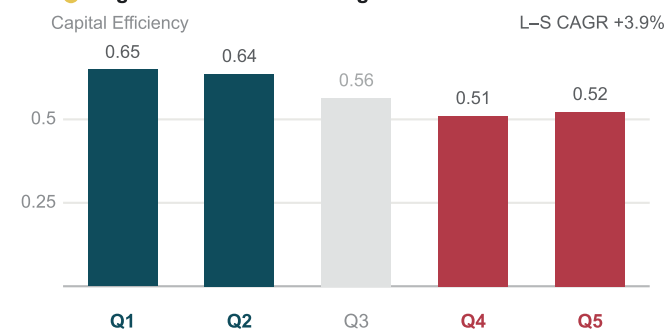
181 ● ROA 60 Month Slope



182 ● Asset Adjusted Capital Investments



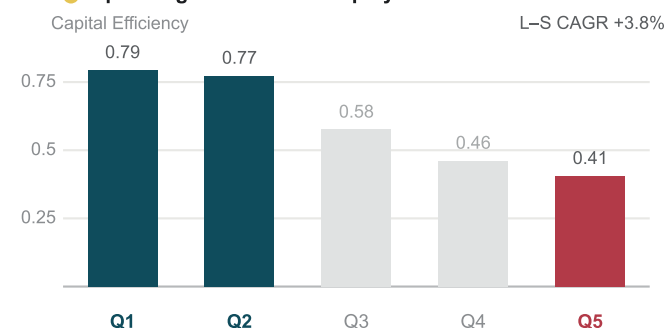
183 ● Degree of Financial Leverage



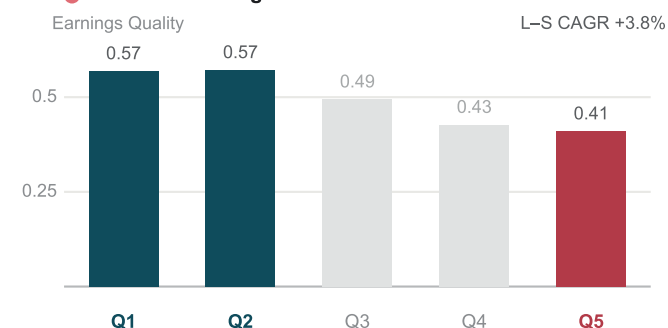
184 ● Return on Equity Stability

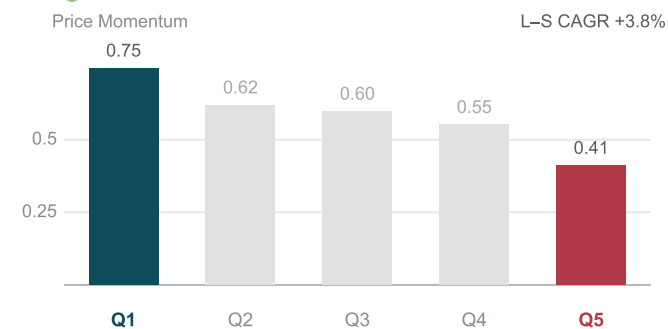
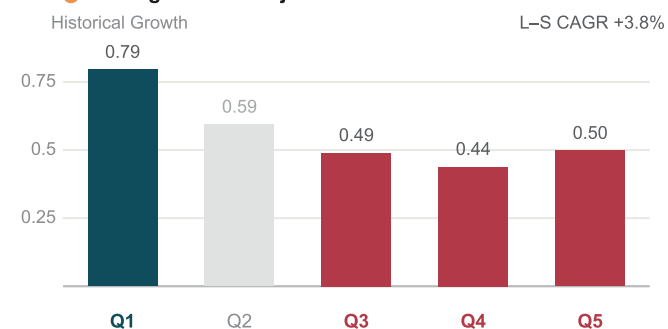
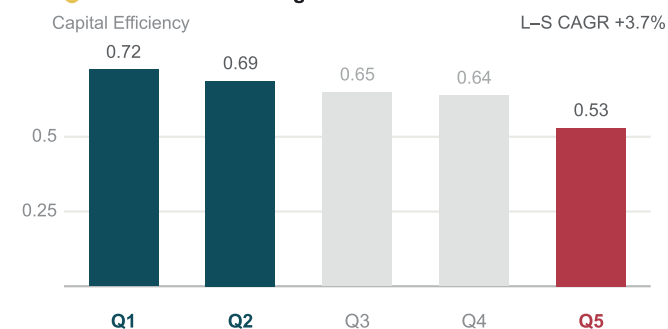
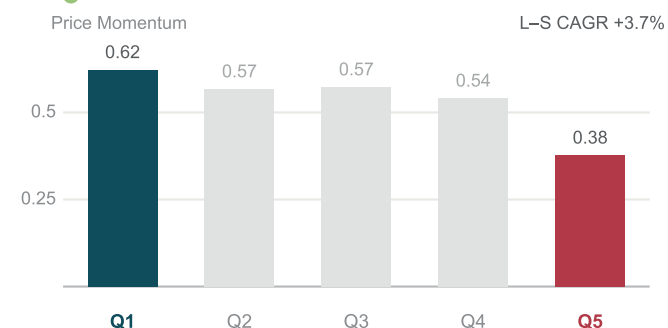
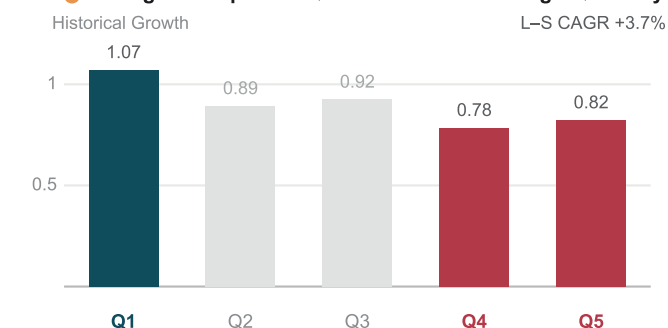


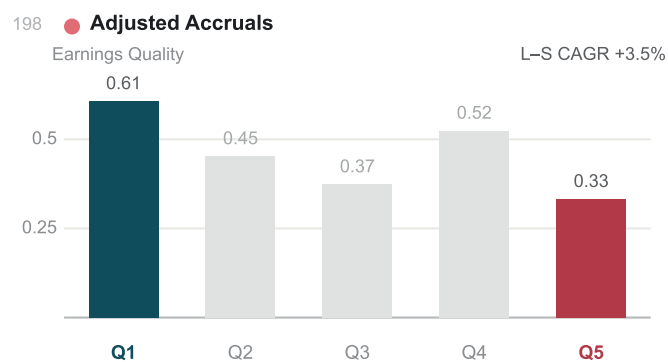
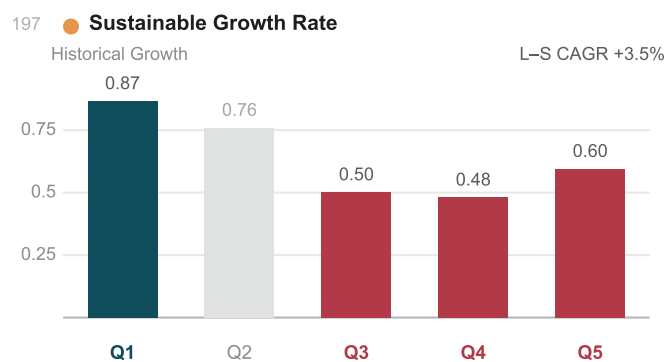
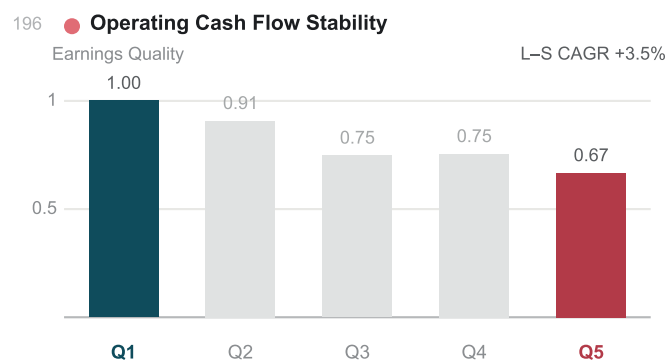
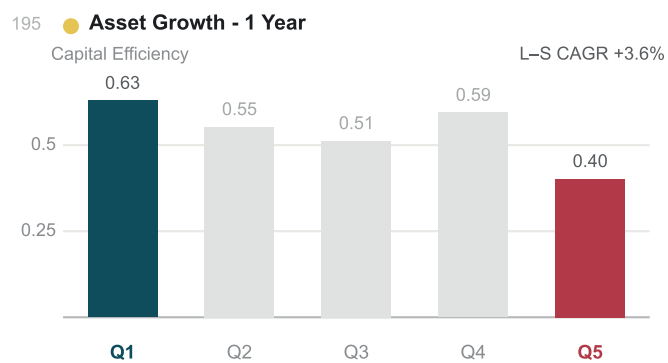
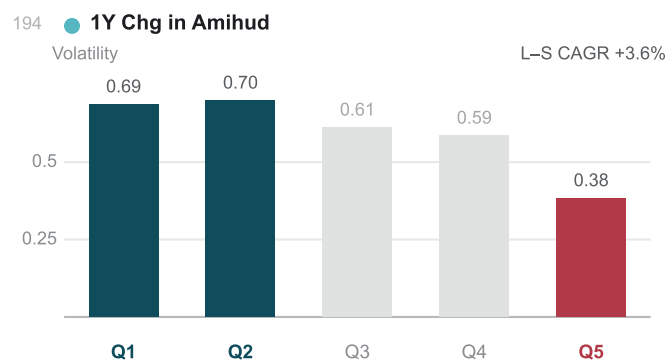
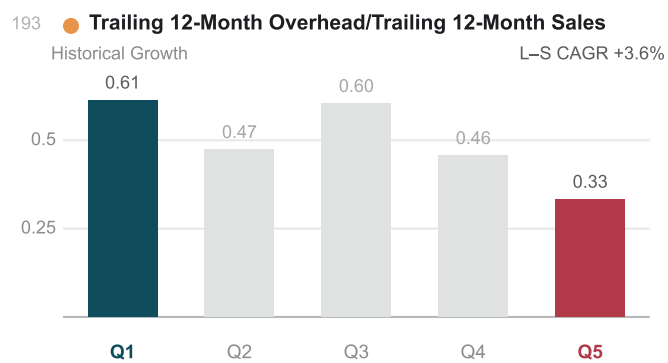
185 ● Operating Cash Flow to Equity

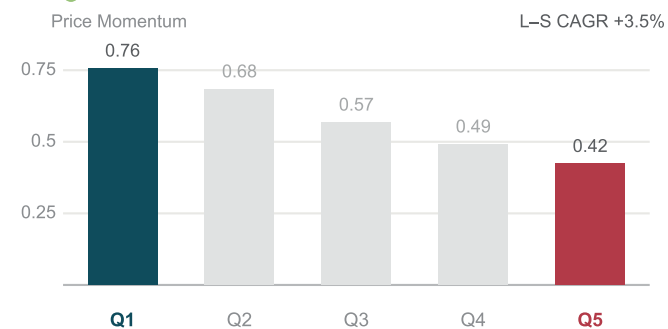
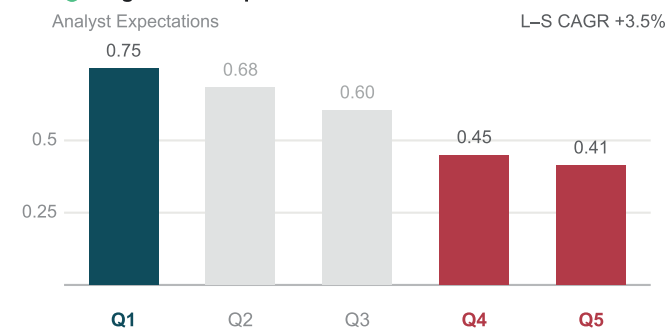
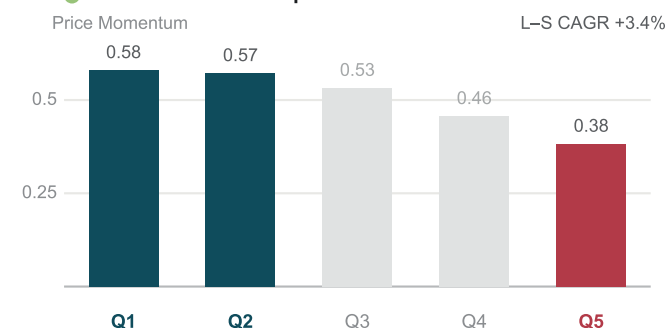


186 ● Retained Earnings to Total Assets



187 ● **Net Profit Margin**188 ● **26W Price Stat**189 ● **3Y Chg in Asset Adjusted EPS**190 ● **Year over Year Change of Total Debt**191 ● **24M Residual Return Variance**192 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**



199 ● **180D Price TStat**200 ● **Target Price Gap to 6M EMA**201 ● **Free Cash Flow to Common Equity**202 ● **1Y Chg in Earnings per Share**203 ● **Working Capital to Total Assets**204 ● **Institution Ownership Concentration**

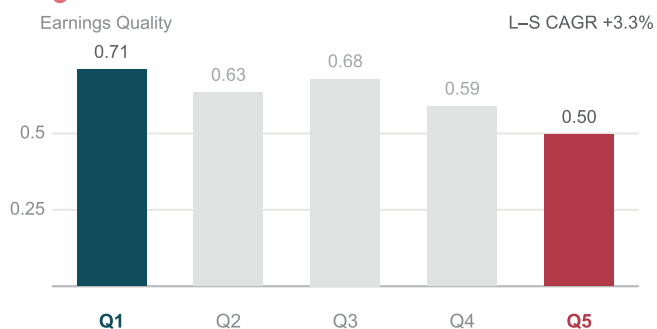
205 ● Return on Assets Stability



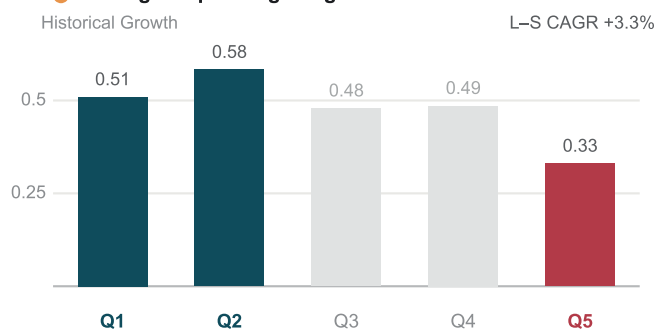
206 ● 3Y Chg in Earnings per Share



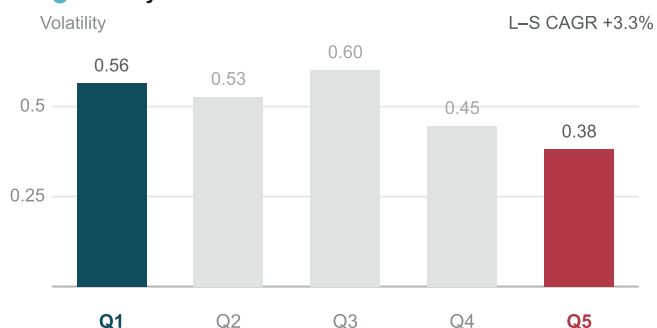
207 ● Accrual Ratio - Balance Sheet



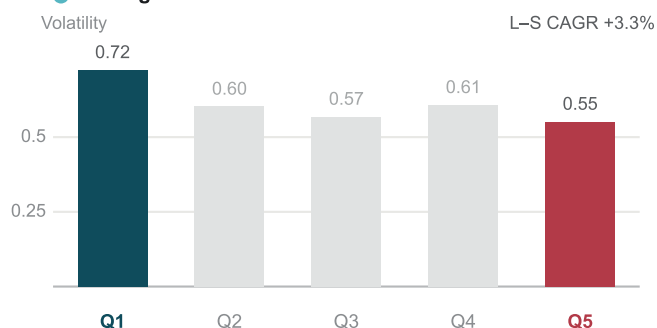
208 ● 3Y Chg in Operating Margin



209 ● 130 Day Minimum Return



210 ● 1Y Chg in Debt to Assets



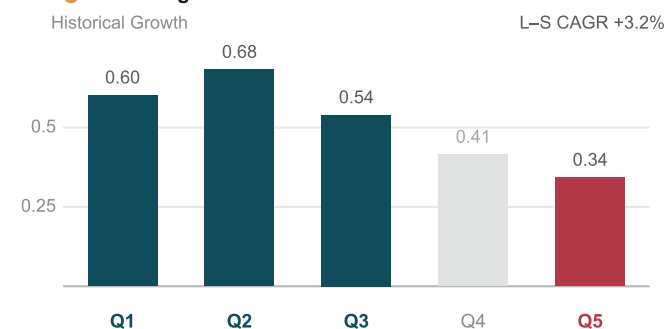
211 ● Ownership Breadth Stability of Institution Ownership by Buyers



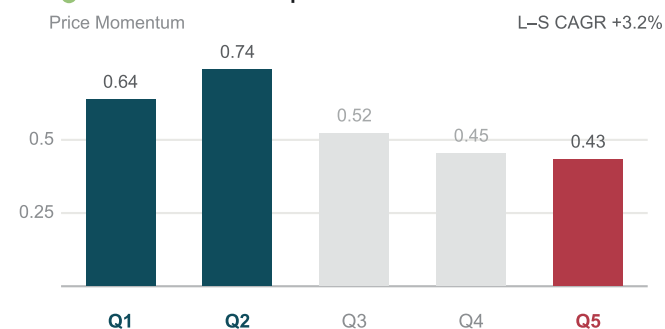
212 ● Pretax Profit Margin



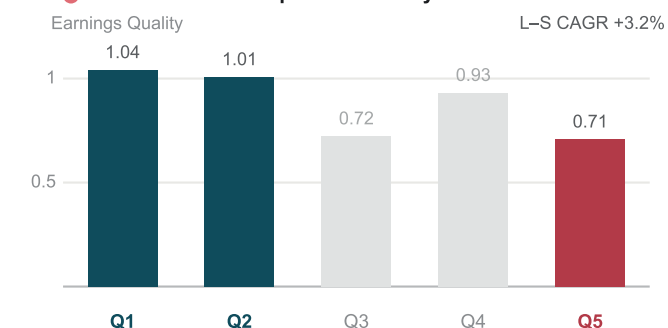
213 ● 3Y Average Annual Sales Growth



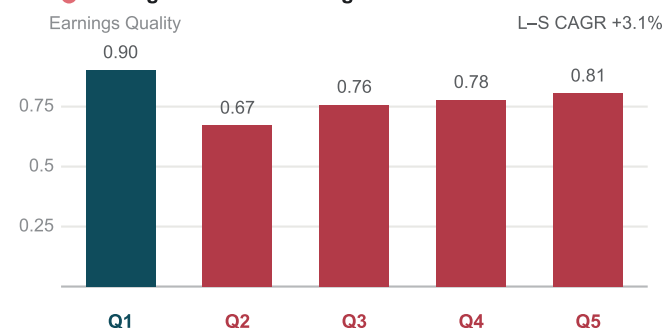
214 ● Institution Ownership Level

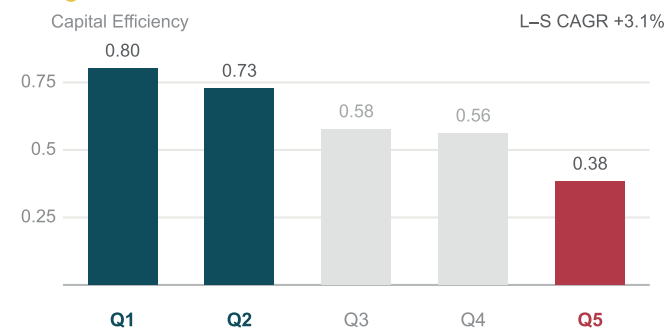
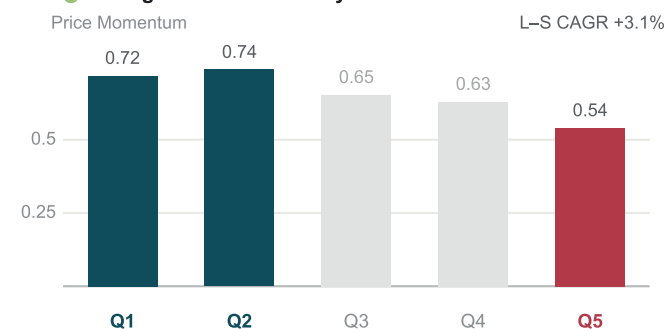
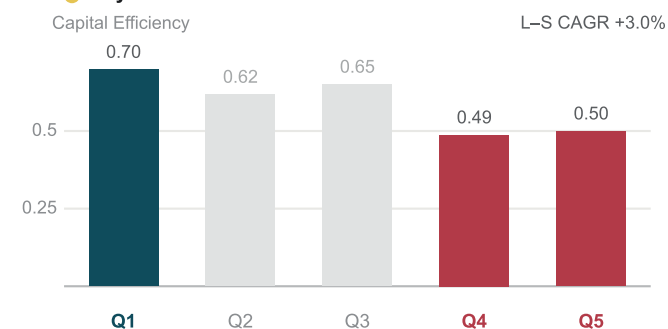
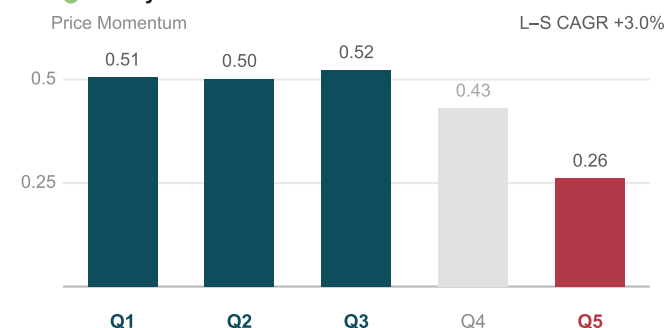


215 ● Research & Development Intensity

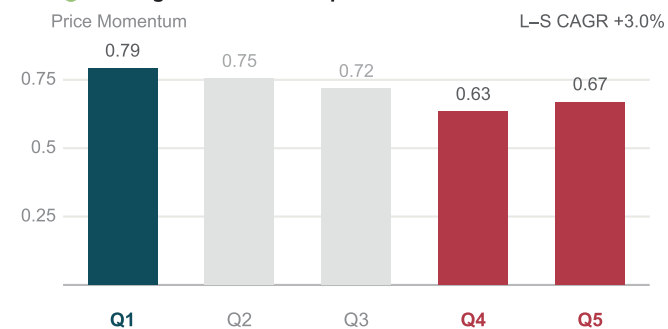


216 ● 1Y Chg in Sales to Earnings

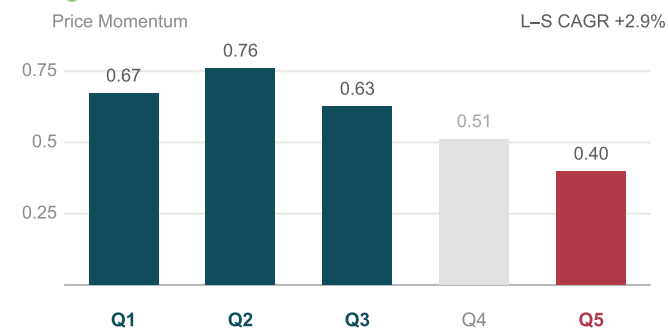


217 ● **Gross Profit to Assets**218 ● **Cash Conversion Cycle**219 ● **Change in Number of Buyers**220 ● **Payout Ratio**221 ● **10 Day MACD Trend**222 ● **Change of Sales to Change of EPS**

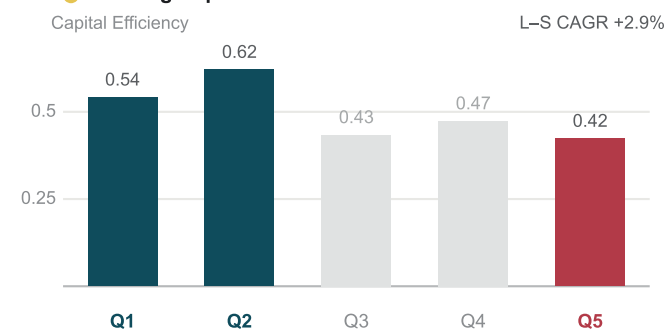
223 ● 6M Chg in 18M CAPM Alpha



224 ● Investment Duration



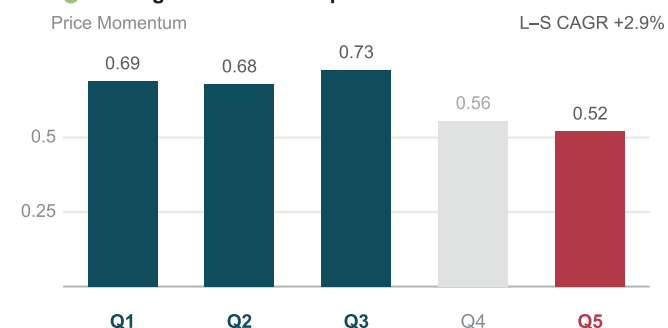
225 ● Working Capital Turnover Ratio



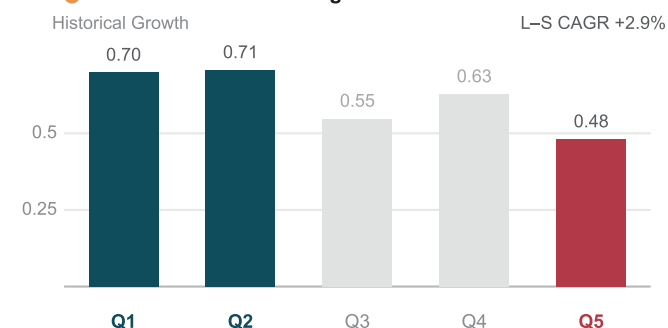
226 ● Ownership Breadth Stability of Institution Ownership by Hol...



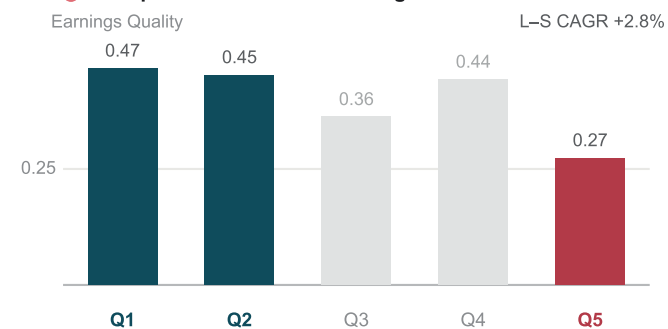
227 ● 6M Chg in 36M CAPM Alpha



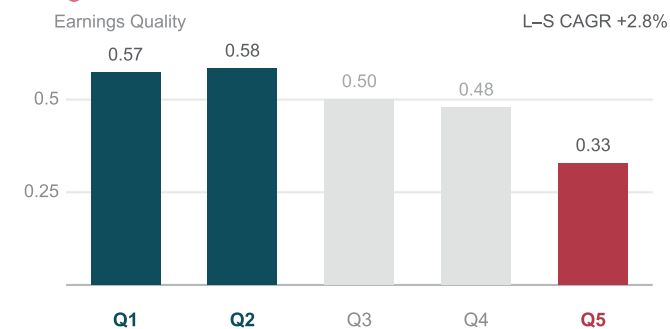
228 ● 3M Momentum in Trailing 12M Sales



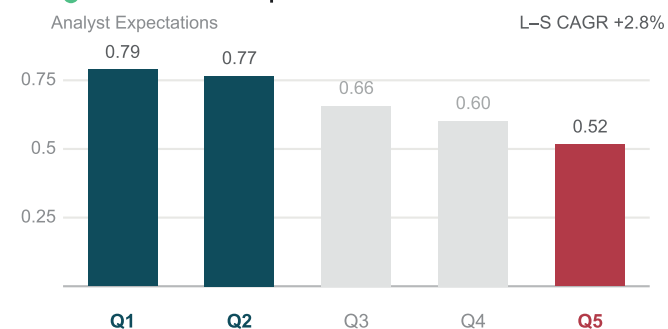
229 ● Unexpected Receivables Change



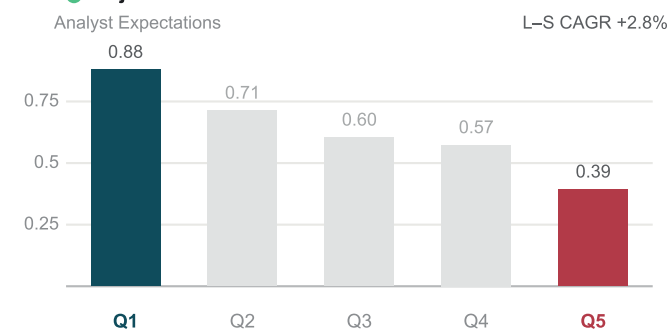
230 ● Cash to Sales



231 ● EBIT Estimate Dispersion



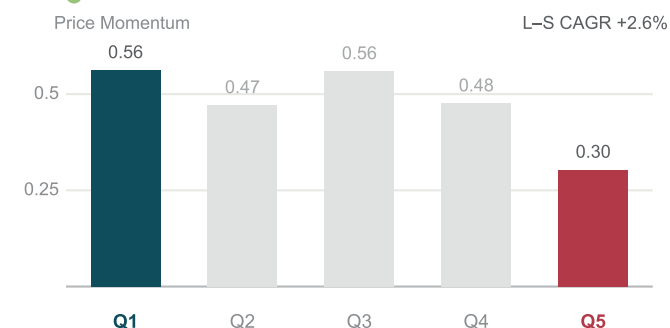
232 ● Adjusted Number of EPS FY1 Revisions



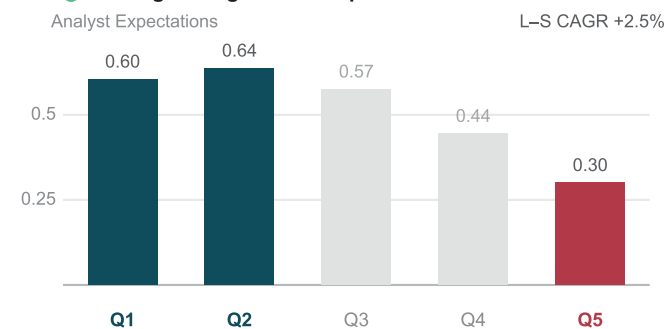
233 ● Volatility adjusted 12M return



234 ● Standardized 36M Residual



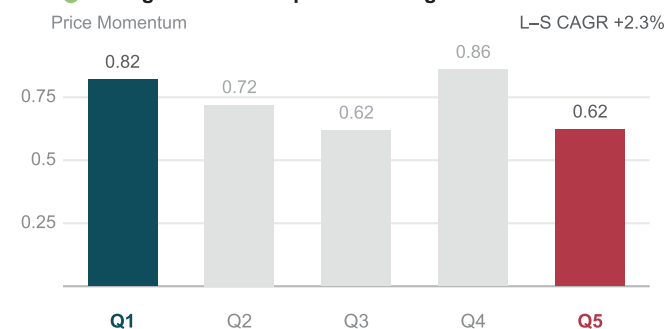
235 ● 6M Chg in Target Price Gap



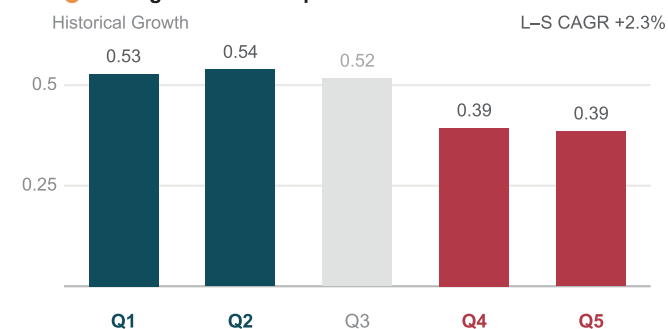
236 ● 1Y Chg in EPS to Operating Cash Flow



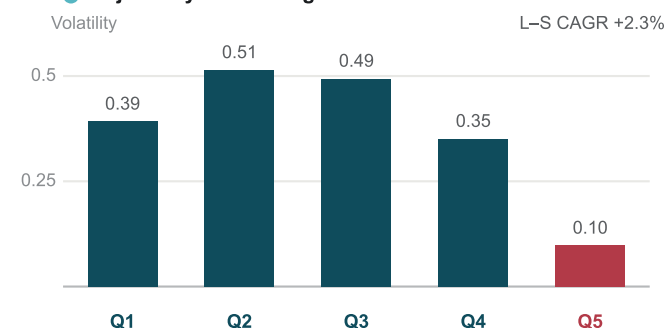
237 ● Change of Ownership Level - Hedge Fund



238 ● 1Y Chg in Cash Flow per Share



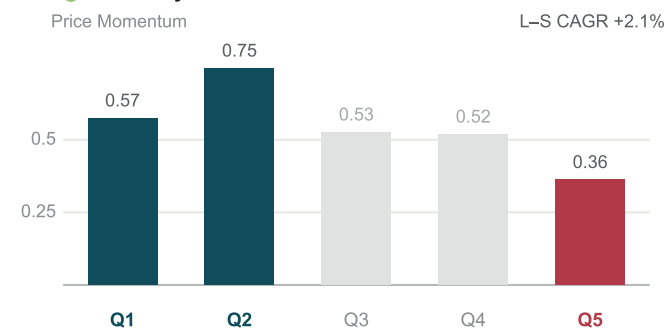
239 ● Adj 50 Day Volume Signal



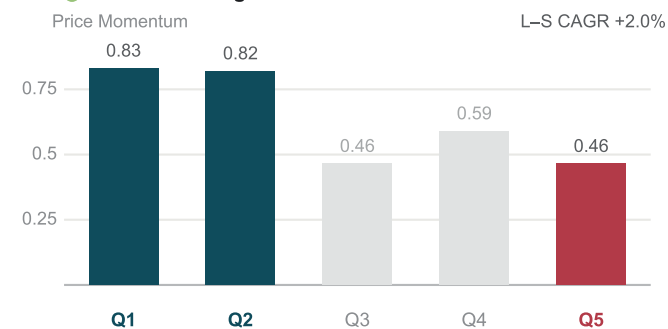
240 ● Analyst Earnings Estimate Diffusion



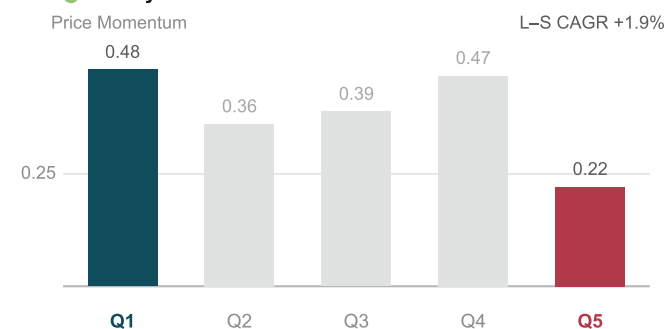
241 ● Max Daily Return in the Past 6 Months



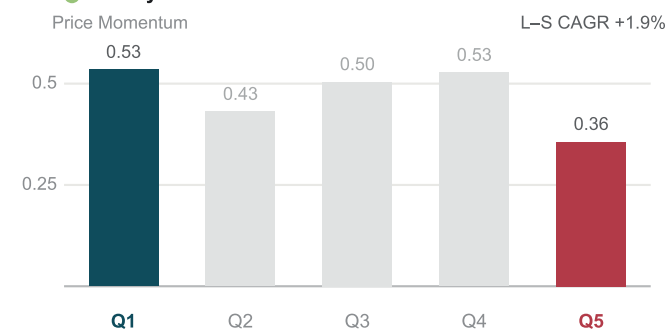
242 ● 6 Month Average Share Turn Over



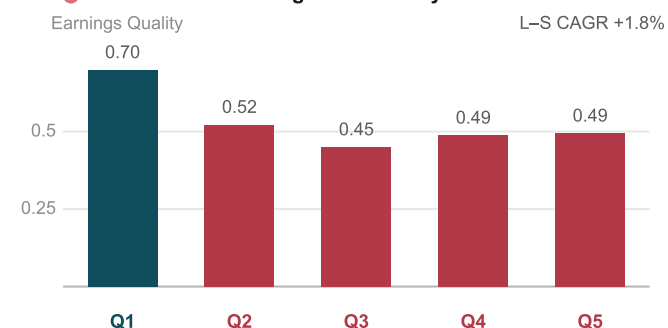
243 ● 30 Day Coefficient of Variation of Volume to Price



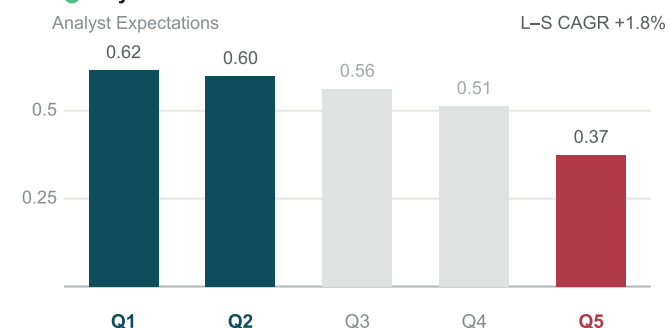
244 ● 20 Day Coefficient of Variation of Volume to Price



245 ● Year over Year Change of Inventory to Assets



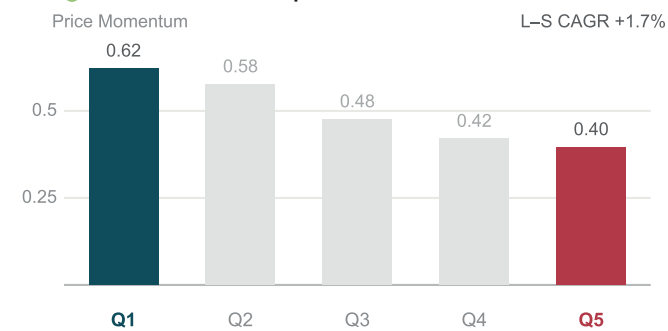
246 ● Buy-to-Sell Recommendation Ratio Less 3M SMA



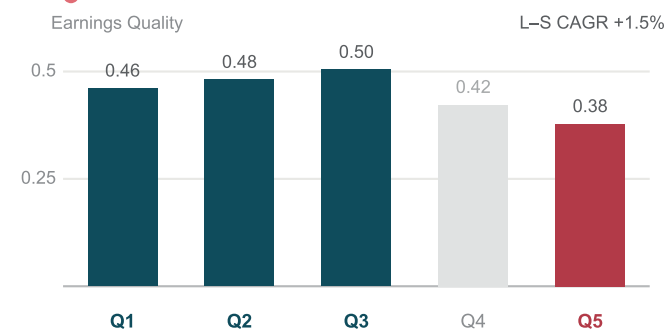
247 ● Closing Price to 52 Week High



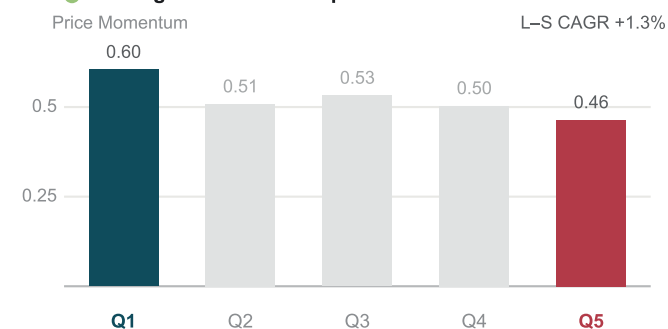
248 ● Institution Ownership Turnover



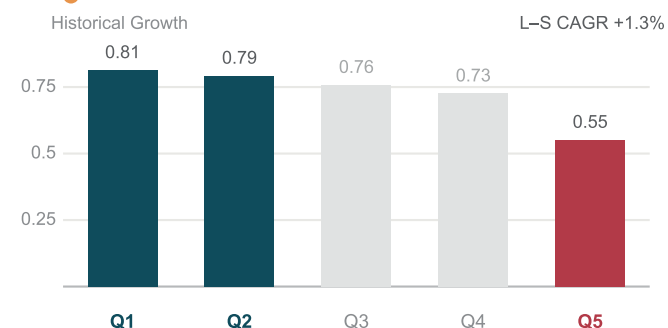
249 ● ROE 20 Qtr Standard Deviation



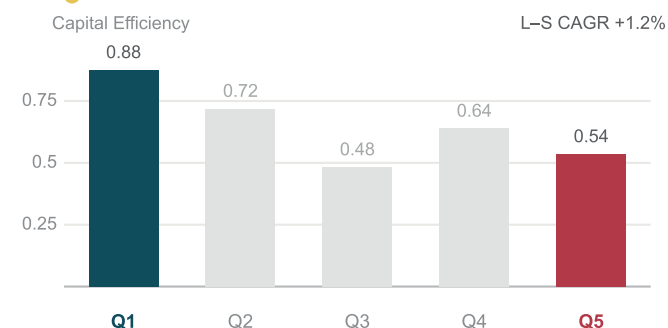
250 ● 6M Chg in 12M CAPM Alpha



251 ● Sales Acceleration



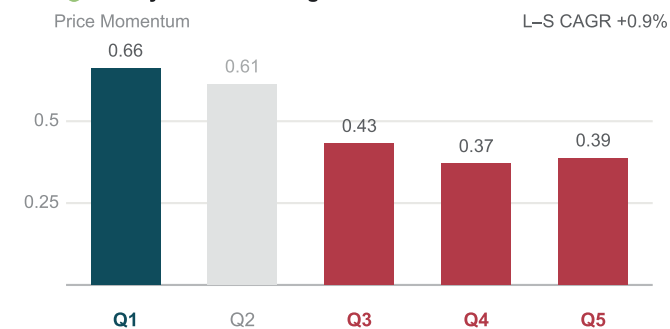
252 ● Normalized ROA



253 ● 6M Avg Chg 1M Recommendation



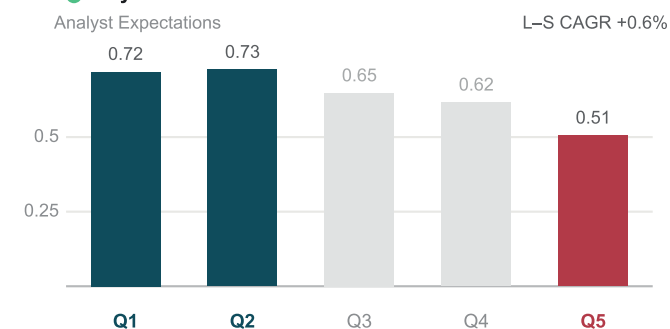
254 ● 14-Day Relative Strength Index



255 ● Change in TTM Depr. to Capex



256 ● Buy-to-Sell Recommendation Ratio Less 3M EMA

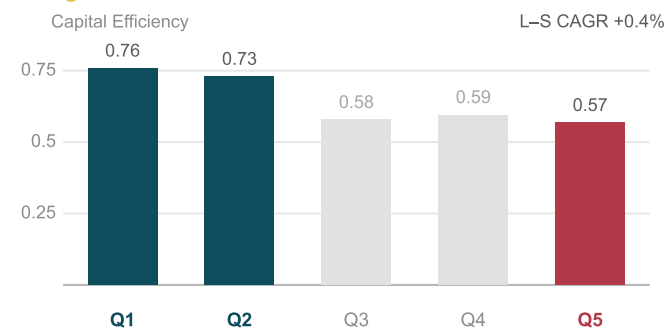
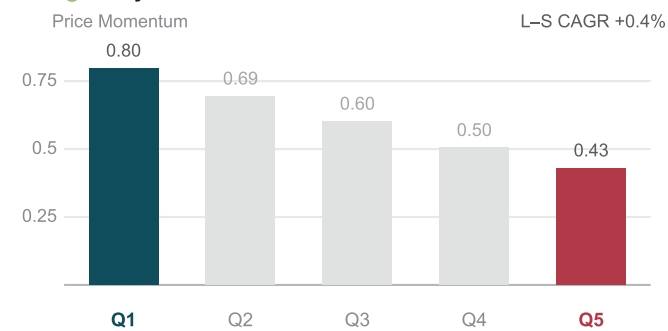
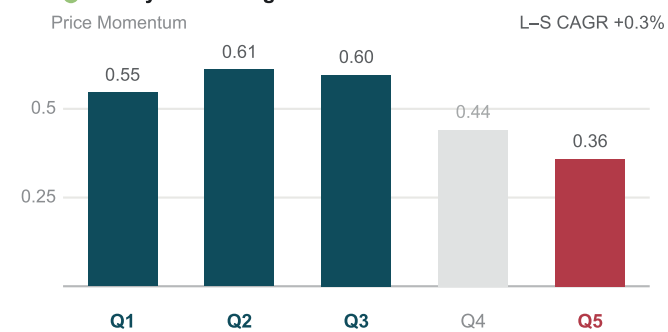
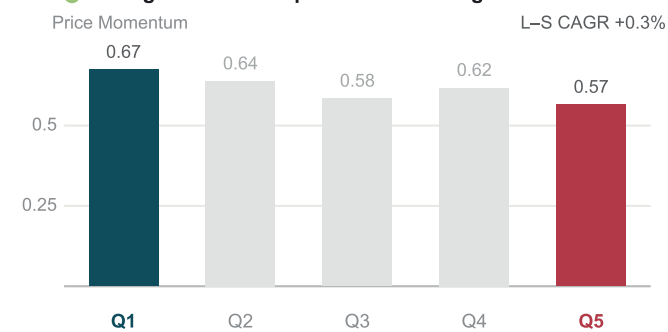
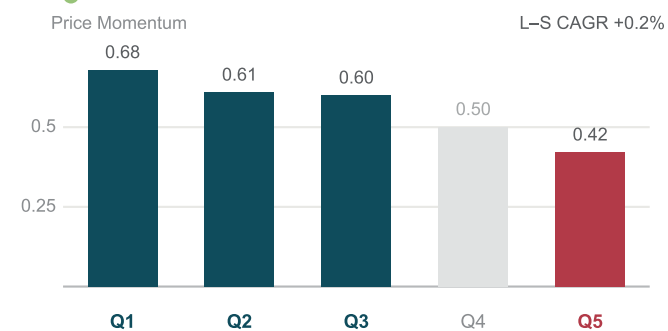


257 ● Revenue Estimate Dispersion



258 ● 8-Week Change in 12-Month Forward Earnings Consensus Estima...



259 ● **Normalized ROE**260 ● **5 Day Price Reversal**261 ● **50 Day Volume Signal**262 ● **Change of Ownership Level - All Managers**263 ● **1M Price Reversal**264 ● **1Y Chg in Share Turnover**